SF EXPRESS 順豐速運

POLYU DESIGN

Improving University Students' Eating Habits with SF Express

Dec 1, 2025

Redesign the Food Ecosystem

Project Goals

Enhance dining efficiency and accessibility

Reduce long waiting time and overcrowding especially during busy hours by providing a smarter, more flexible meal-ordering and pickup experience.

Promote healthy eating habits among students

Encourage balanced and nutritious food choices through personalized recommendations and visibility of meal information, such as calorie content and nutritional value.

SF EXPRESS 順豐速運

POLYU DES*GN

SF EATS 豐食

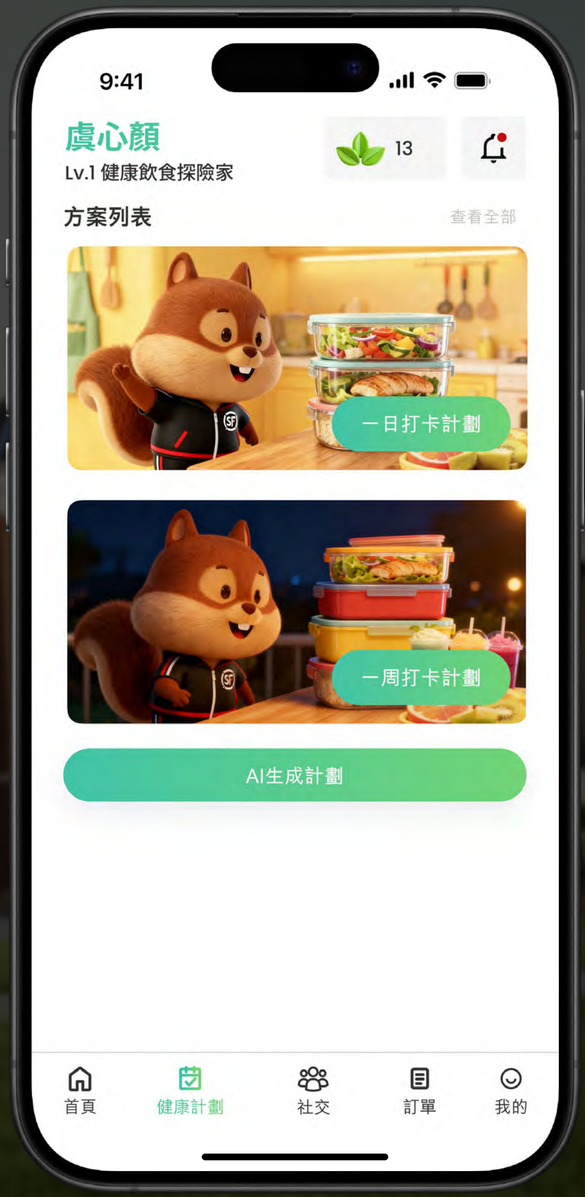
A food delivery app that brings you healthy meals and tailors a personalized diet plan.

Group Order

Pre-Ordered Meals

Community

AI Diet Plan



SF EXPRESS 順豐速運 POLYU DES'GN

SF Eats
Mascot

Speed & Precision

Squirrels move fast, darting around with precision.

- Quick delivery
- Responsiveness
- Efficiency

Foraging = Food Discovery

Squirrels are natural foragers. They hunt, gather, and store food — which can translate into:
Helping users finding the perfect meal and delivering what people crave.

Friendly & Energetic

A squirrel is friendly and energetic, which is perfect for our target audience - university students.

The Community
of SF Eats

"Your journey to better health isn't a solo path anymore."

SF Eats

Group Order & Events

Group Order Function

Our **group order** feature lets students combine meals into one shared order, saving time and delivery costs while making healthy eating easier and more social for study groups, classmates, or friends.

Weekly Events to Gather Users

Our healthy food delivery app offers weekly themed events—like **Green Monday** with plant-based meals—to make healthy eating fun, varied, and easy for students.

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AI-Powered, Personalized Diet Plans

& Pre-Ordered Meals

Students can enter basic details—such as height, weight, and meal budget—to enable the AI to create a **personalized diet plan**. This plan encourages healthier eating habits and helps students sustain them over time.

Additionally, the service assists students in choosing and pre-ordering meals that align with their personalized plan.

SF Eats

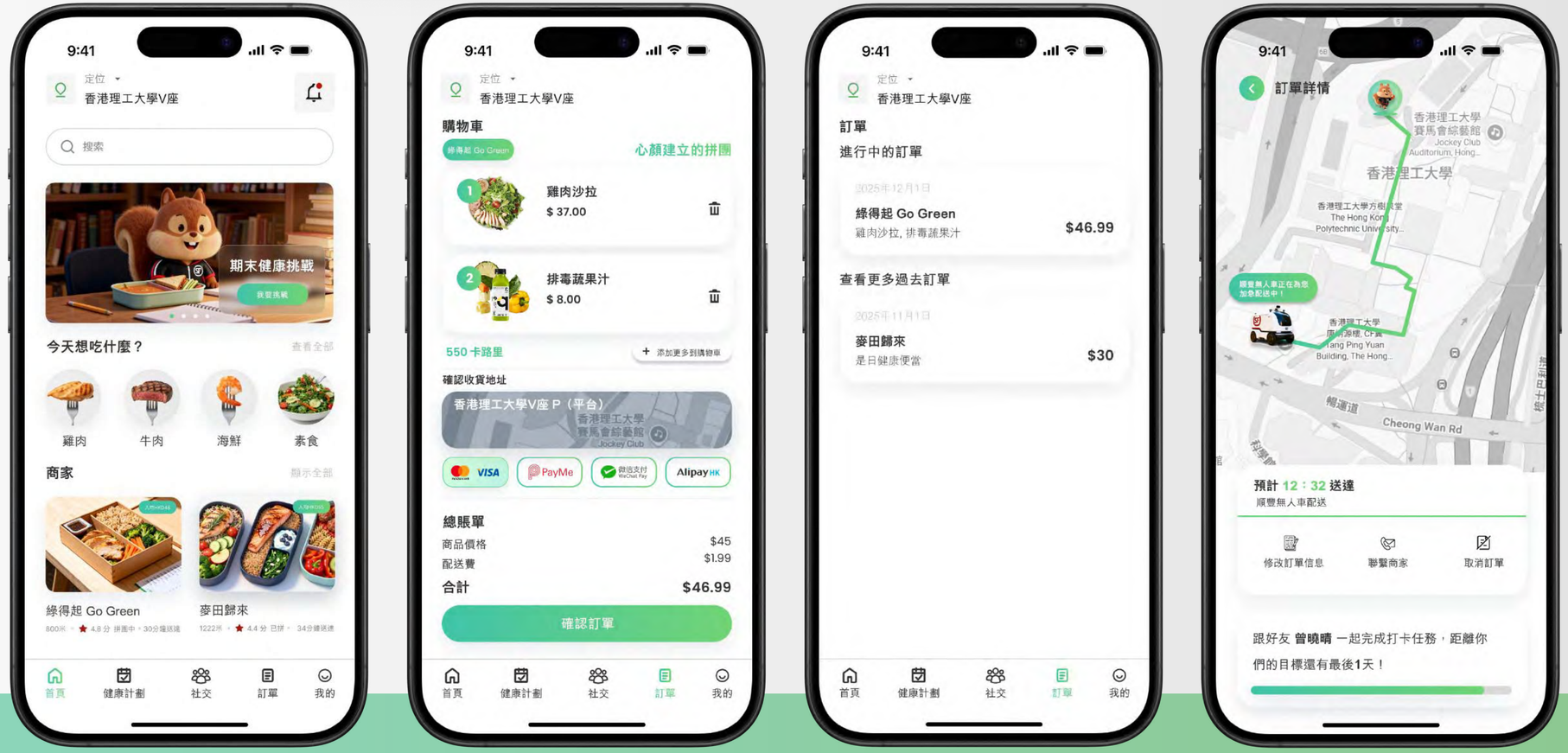
Ordering Process

We aim to provide students with a **seamless, efficient** food delivery experience powered by SF Express' unmanned vehicles.

3 Options

- AI Meal Recommendations
- Food Picking by User
- Pre-Ordered Meals

→ Shopping Cart → Ordering Takeout → Delivery Tracking



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SF Eats Challenges & Streaks

From AI Diet Plan to Streaks

Users can maintain their streak by consistently following their AI-generated diet plan. This feature is available only to users who choose to participate in the AI diet program.

SF EXPRESS 順豐速運

POLYU DES'GN

點餐順，食飯順 ，豐食都順。

Order smooth, eat smooth, SF eats is smooth.

Enhancing Customer Experience of Bambu Lab's 1st Flagship Store

Untitled Fridays

Oct 10, 2025

Project Goals

Enable **first-time visitors** to become **confident makers**.

Connects visitor imagination with the 3D-printed displays.

Ultimately lead user into Bambu's service ecosystem.

- Turn Curiosity into Capability
- Creating a Cohesive User Journey
- Lead Generation & Sales Conversion
- Brand & Ecosystem Promotion

Bambu Lab POLYU DES°GN

Co-Design Workshop

3 Sections

From Pain Points to Opportunities

We asked participants to write their pain points on sticky notes based on the store's three main problem areas: Display, Guidance, and Sandbox.

Inspirational Case Sharing

We invited participants to share inspiring examples of service design or experience design.

Co-Creation

Using the Bambu Lab store's 3D model, participants redesigned the entire store layout and reimagined the customer journey.

Bambu Lab POLYU DES'GN

Models Display Wall

Workshop Space

Sandbox

Photo Wall

Bambu Market

Display Wall

3D Printer Sell Area

Filament Display Wall

Floorplan – After

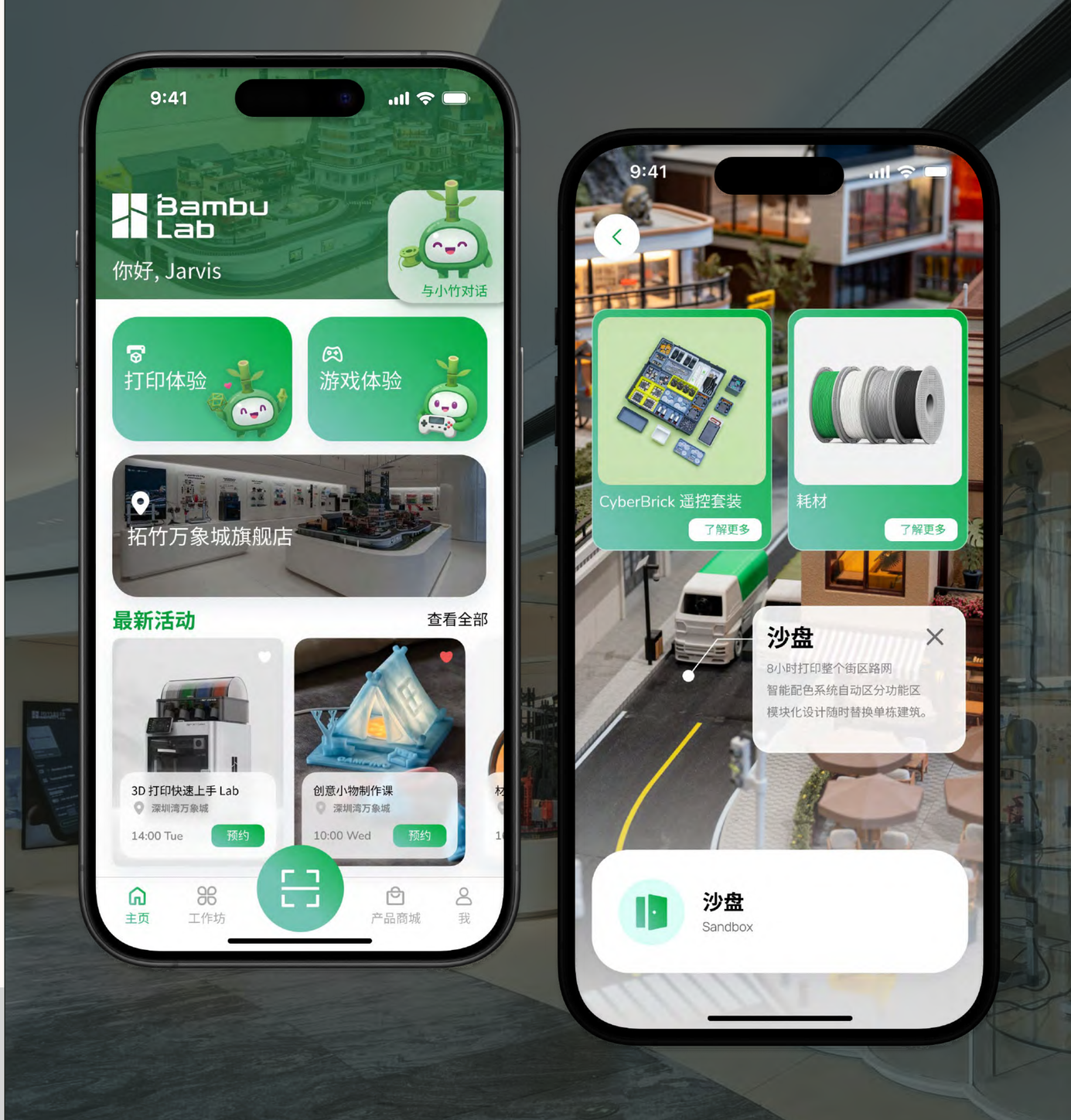
New Customer Journey

Bambu Lab POLYU DES·GN

Mini Program
Design Concept

A redesign of the Bambu Lab mini program, with easy to use and interactive interface.

Scan the QR code and try out our mini program!



Little Bambu 小竹 AI Chatbot

Hi~ I'm Little Bambu! Feel free to ask me anything, and I'll do my best to answer it!

Main Features

AR Guided Tour

Product Store

Workshops Reservation

3D Printing Service Reservation

Key Strategic Outcomes

Immersive AR feature attracts users and makes complex technology **accessible**.

Workshop booking transforms customers into a skilled community, increasing **product adoption** and **loyalty**.

By providing an superior, all-in-one experience, from learning to buying to printing, and foster a **loyal ecosystem**.

By educating users to reduce churn rates, which directly **increases consumables sales** and **user loyalty**.

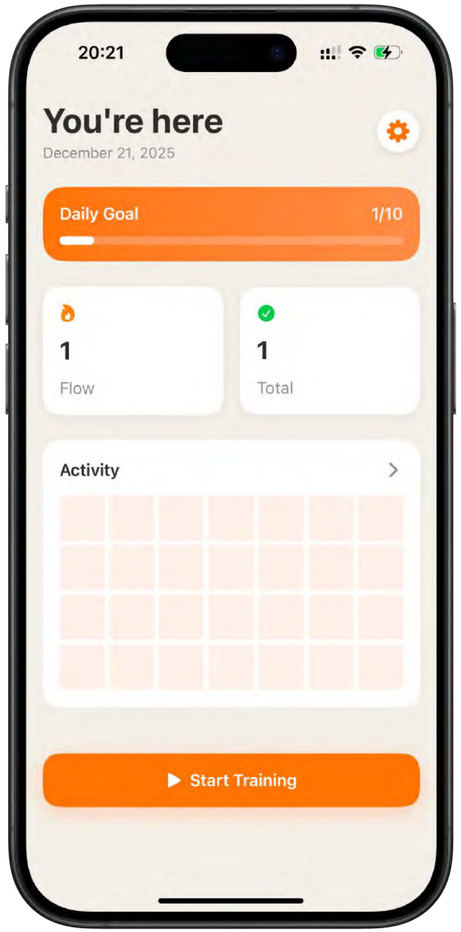
Just A 2nd

Sharpen your ears in seconds

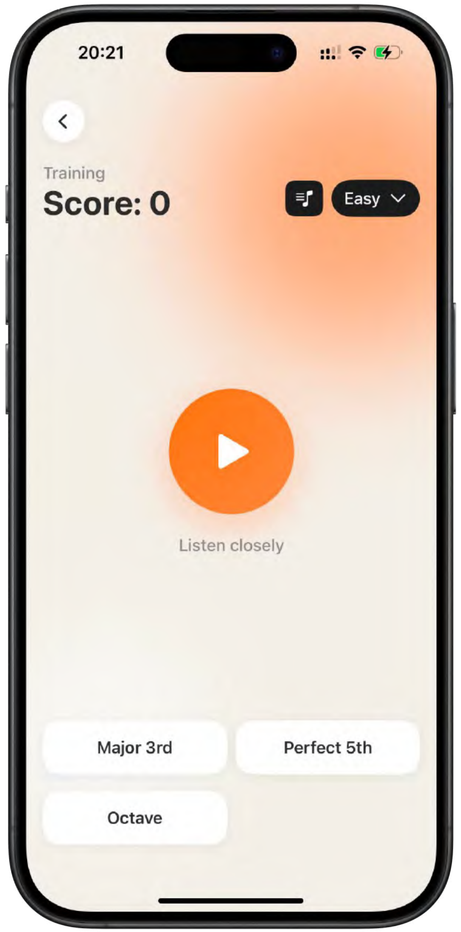
Say goodbye to boring music theory drills! Whether you are learning a new instrument or want to understand the music you love, Just A 2nd is your new pocket-sized trainer.

Spend a few seconds a day and improve your hearing skills!

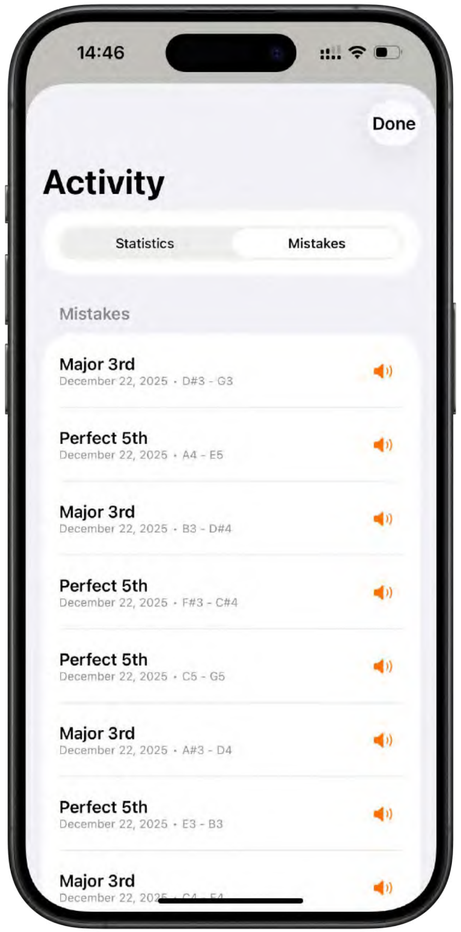
Download on the App Store



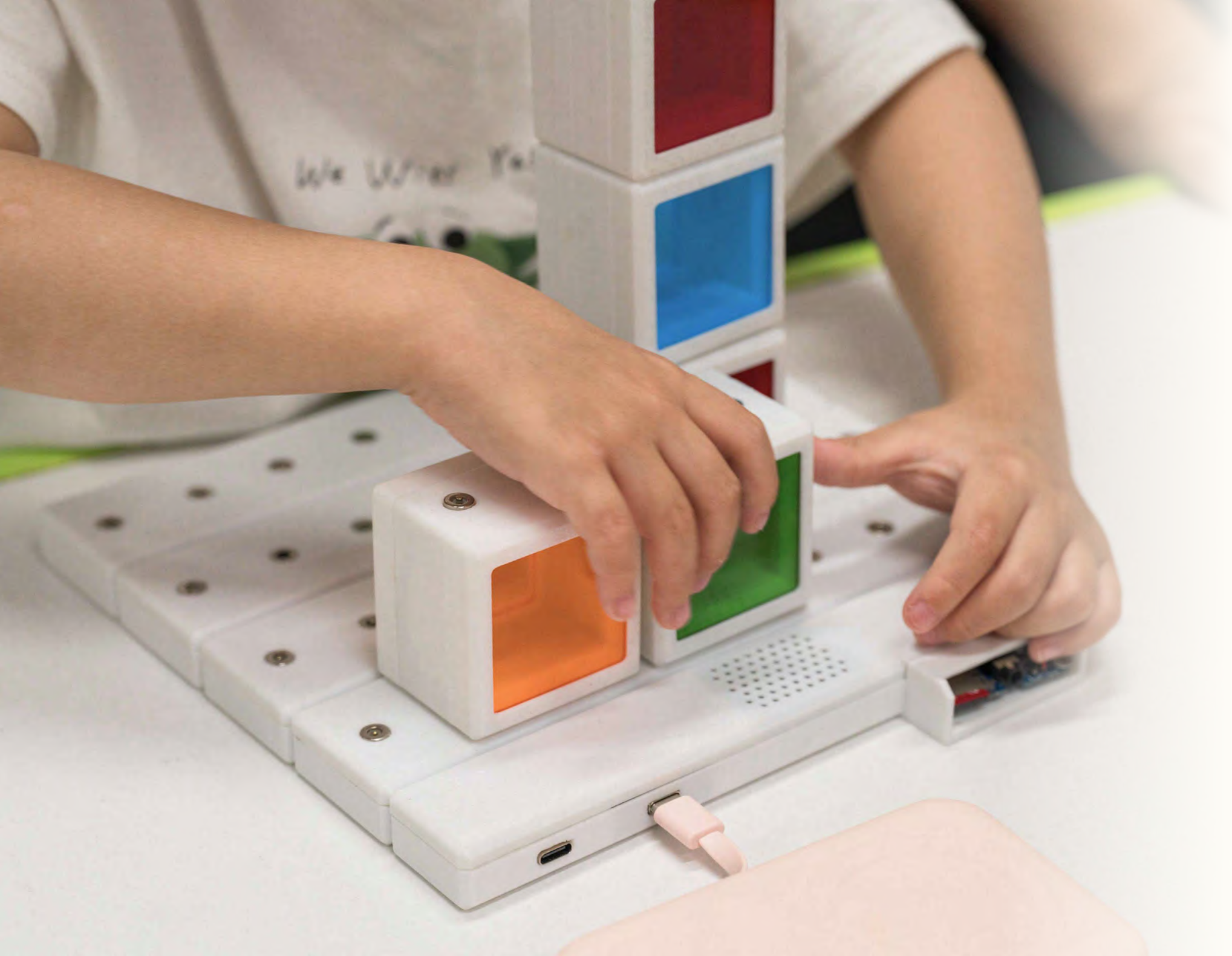
Stay motivated with goals and daily streaks.



Master musical intervals with focused listening exercises.



Learn from mistakes.



01

SoundBricks

A SSP sound interactive toy for ASD kids

Individual Project

Making-Of: https://youtu.be/G_39gM8YMLA

Brief

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that typically manifests in early childhood. Intervening in the repetitive behaviors of children with autism is crucial. Existing research suggests that sensory stimulation reinforcement or extinction methods can effectively intervene in these behaviors.

Key Words

ASD Kid's Toy | Stimulus-Stimulus Pairing | Interactive Design

Ideation & Concept

SoundBricks employs Stimulus-Stimulus Pairing (SSP) to intervene the repetitive behaviors of children with autism. SoundBricks use sound as a neutral stimulus and combines it with the interactive experience of building bricks, providing children with continuous sensory stimulation during play. This aims to reduce repetitive behaviors while enhancing their spacial imagination and fine motor skills. This toy not only offers a fun and engaging platform for children with autism but also provides an innovative and effective intervention tool for parents and educational institutions.

Background Research

About Autism Spectrum Disorder

A neuro-develop mental disorder

Autism can manifest very differently in each person. For example, some are nonspeaking, while others have proficient spoken language.

Affects every aspect of a person's being

how they perceive, experience, interact with and interpret the world

A minority neurology but affect a llarge number of people

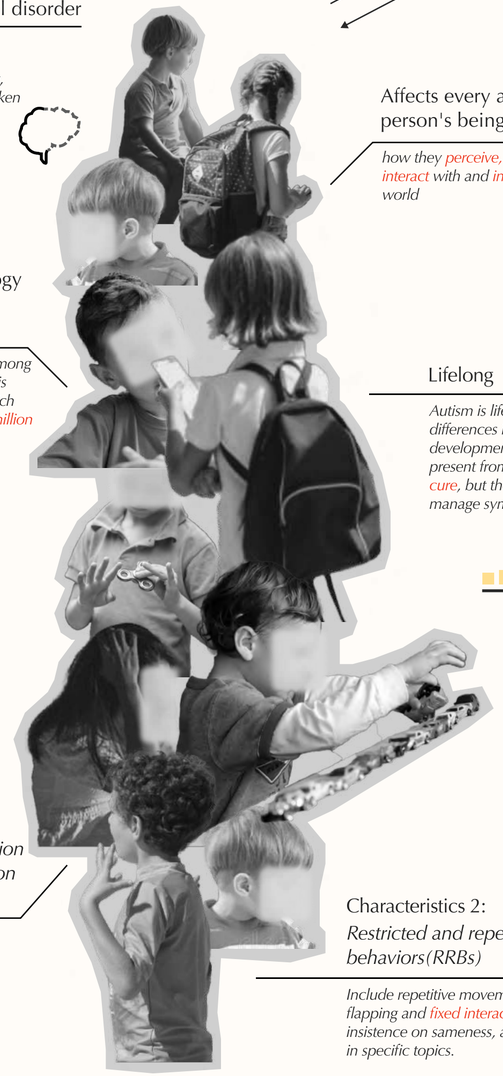
The prevalence of ASD among children aged 6-12 years is approximately 0.7%, which translates to about 1.73 million children.

6 — 12

0.7% - 1.73 million

Lifelong

Autism is lifelong due to differences in brain development and genetics present from birth. There's no cure, but therapies can help manage symptoms.



Characteristics 1:

Defects in social interaction and social communication skills

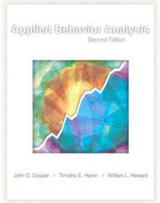
Include difficulties with making eye contact, interpreting nonverbal cues, engaging in reciprocal conversations, and forming relationships.

Characteristics 2:

Restricted and repetitive behaviors(RRBs)

Include repetitive movements (like hand-flapping and fixed interaction for toys), insistence on sameness, and intense interests in specific topics.

Applied Behavior Analysis (ABA)



Cooper, J. O., Heron, T. E., & Heron, W. L. (2020). Applied Behavior Analysis(Second Edition).

Applied Behavior Analysis (ABA) is a therapy that uses principles of learning and behavior to improve skills in areas like communication, social interactions, and academics. It's often used for individuals with autism.

The ABC Model

The main principle of ABA is a behavior chain. Behavior chain includes the antecedent event, behavior, and the consequence.

a directive or request to perform an action

Antecedent

- Prompting
- Goals
- Feedback

Behavior

a response for antecedent

- Undesired
- Desired

Consequence

a reaction from the therapist

- Positive Reinforcement
- Negative Reinforcement
- Punishment
- Extinction
- Escape/Avoidance

Stimulus - Stimulus Pairing (SSP)

Respondent conditioning (classical or Pavlovian conditioning) experiment. - Ivan Petrovich Pavlov (1849–1936)

Before → *After - Stimulus - Stimulus Pairing*

Conditioned stimulus	**Neutral stimulus**
Food	Metronome
Salivation	No Salivation

Food + Metronome → Pairing → **Conditioned reflex**

sound of metronome followed by presentation of food → Salivation

Neutral stimulus Metronome → **Unconditioned stimulus**

Similar effect as conditioned stimulus (food)

Example of SSP Intervening RRBs

Li Yan. (2019). The Intervention of Stimulus - Stimulus Pairing On The Restricted and Repetitive Behaviors of Autism Children.

Individual Cases

Case 1: H, Male, Age 5

The attention to the picture is not bad, and the attention is not focussed enough.

Restricted Behavior: Press fingers and shake body.

Case 2: C, Male, Age 4

Low concentration, low degree of cooperation, frequent out-of-seat behaviour.

Restricted Behavior: Walking on tips, curling up on the floor, looking at hands

Case 3: Z, Male, Age 4

Good rigid memory ability, average attention, and average coordination.

Restricted Behavior: Frequently open and close the door

Experimental Procedure

Stage One (Pairing)

Show the Toys → Delay for 3s → The individual can operate the toy 5s without ant assistance → Primary and social reinforce

Unexpected response (Prompt Stage): Gesture / Demonstration / Physical prompt

(From top to bottom, step by step)

Stage Two (Detect)

Show the Toys → Delay for 3s → The individual can operate the toy 5s without ant assistance → Mark as (+)

Unexpected response → Mark as (-)

The Effectiveness of Intervention

Significant Instant Effect

Effectively reduce the RRBs of autistic children during the experiment.

Good Maintenance effect

The effect of SSP can be maintained for a period of time after the intervention, showing a good long-term effect.

High social validity

Parents and teachers highly appreciate the intervention effect of SSP.

Field Research

Child Development Center in Hangzhou

In order to better understand these children, I visited a Development Center in Hangzhou, China. The centre has served more than 300 special children, of which 70% have successfully entered kindergartens and primary schools.

Interviews With Teachers

Ms. Jin, Institution Manager/Multi-field Teacher

Q: What will experience if a kid would like to go to the institution?

A: Parent perception - hospital consultation - institutional parent consultation - child education assessment - enrolment - formulation of curriculum plan IEP - insist on continuous and effective intervention (adhere to intervention, home-school cooperation)

Mr. Liang, Sensory Integration Teacher

Q: What's the most important age for intervention?

A: The sooner the better. Before the age of six we call it rescue intervention. Presently, the education level improved, and the diagnosis of children is advanced, we have kid only 12 and a half months in the institution.

Q: How to know if a kid could be autism?

A: If the kid has cognitive backwardness (don't know cake/water), bad understanding and couldn't express clearly... You should be careful.

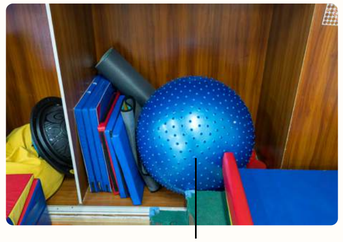
Types Of Toys Commonly Used

01/ Foam Fruits

Cognitive competence
Social skill
Command capability

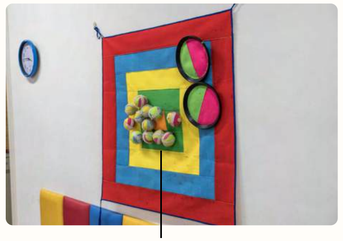
02/ Geometry Block Toy

Shape cognition
Hand-eye coordination
Fine motor skill



03/ Sand hammer

Auditory stimulus
Rhythm sensation



04/ Yoga Ball

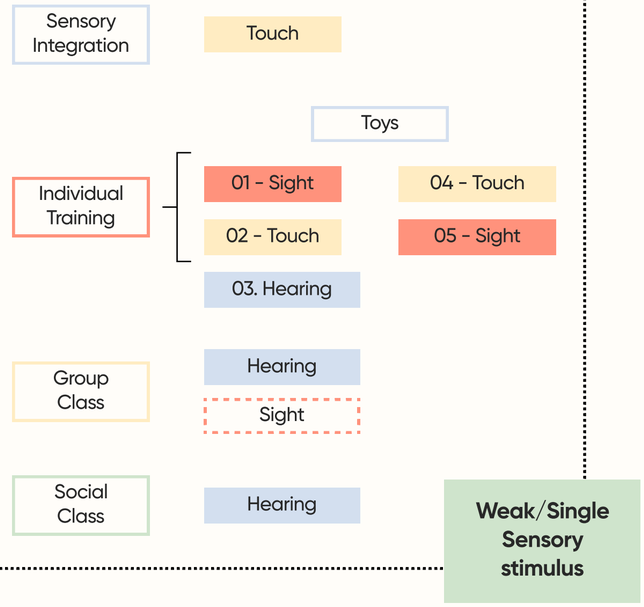
Gross motor skill
Improve balance
Core muscle training

05/ Sticky Ball

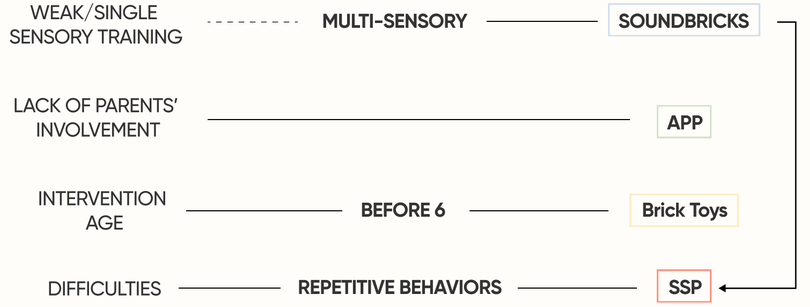
Color cognition
Hand-eye coordination
Fine motor skill

The institute use a variety of toys. There are cases where a toy can exercise multiple abilities in different contexts. A open toy is easier to fit different kids.

Training Analysis



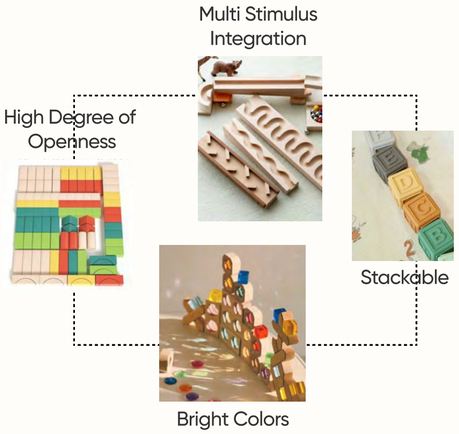
Conclusion & Ideation



The current trainings are single or weak multi-sensory trainings. SoundBricks will target the ASD kids with age under 6, and try to integrate sound, visual and physical stimulus together to intervene RRBs. The APP will be a digital interface to the toy and allow parents to take part in the intervention.

SSP on Brick Toys

The Appearance Concept

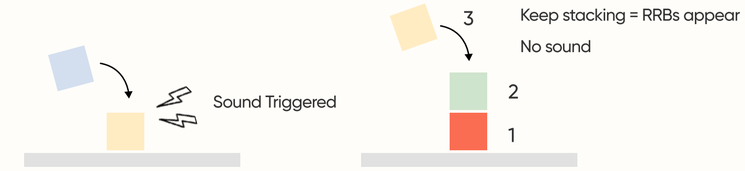


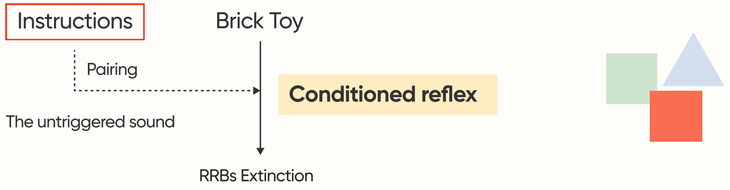
Brick toys should be suitable to contain multi stimulus and easy for kids under 6 to play.

The ASD kid is easy to be attracted by bright colors, transparent acrylic is used as visual stimulus.

Interaction

Sensory stimulation reinforcement or extinction methods can effectively intervene in stereotyped behaviors.



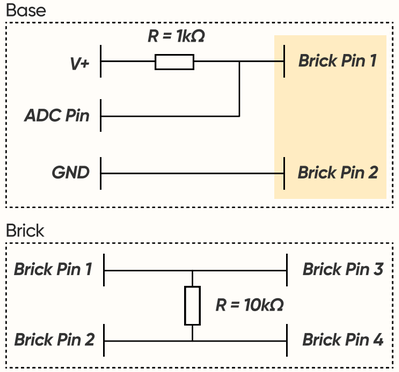


By pairing neutral stimulus with unconditioned stimulus, repetitive behaviors can be intervene.

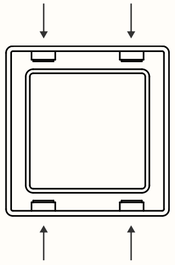
Prototyping

Making-of: https://youtu.be/G_39gM8YMLA

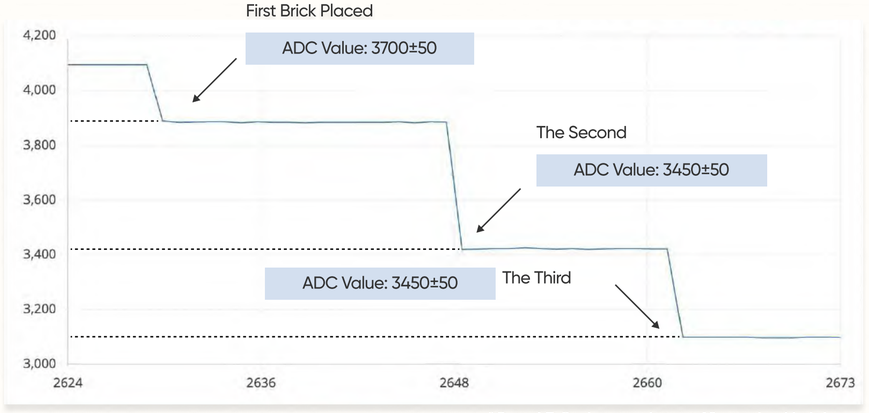
Circuit Design



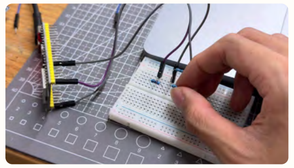
Quick-separable Magnet

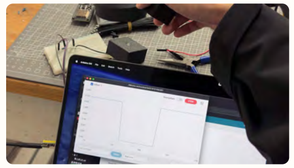


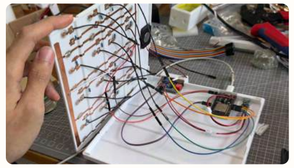
These contacts allow bricks to be connected and separated quickly and solidly, which is the key to get a stable ADC readout.



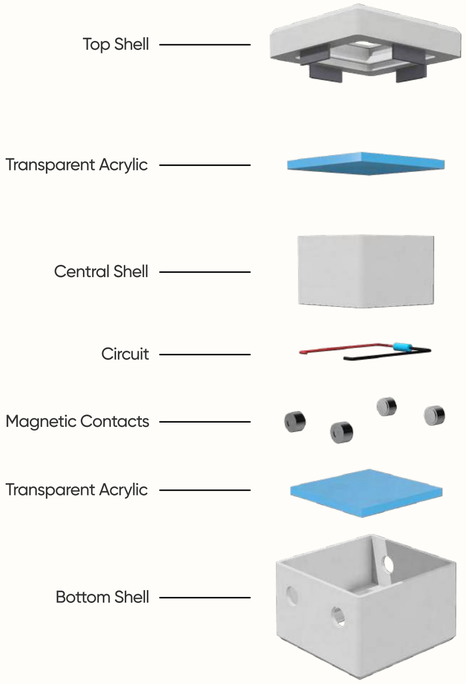
*One ADC pin can detect 10 stacked bricks







Explode View



*Acrylic is inserted while printing

APP Design

APP Control Procedure

Device on → **Sounds Selecting** → **Pattern Generate** → **Build the Pattern** → **Share the Play**

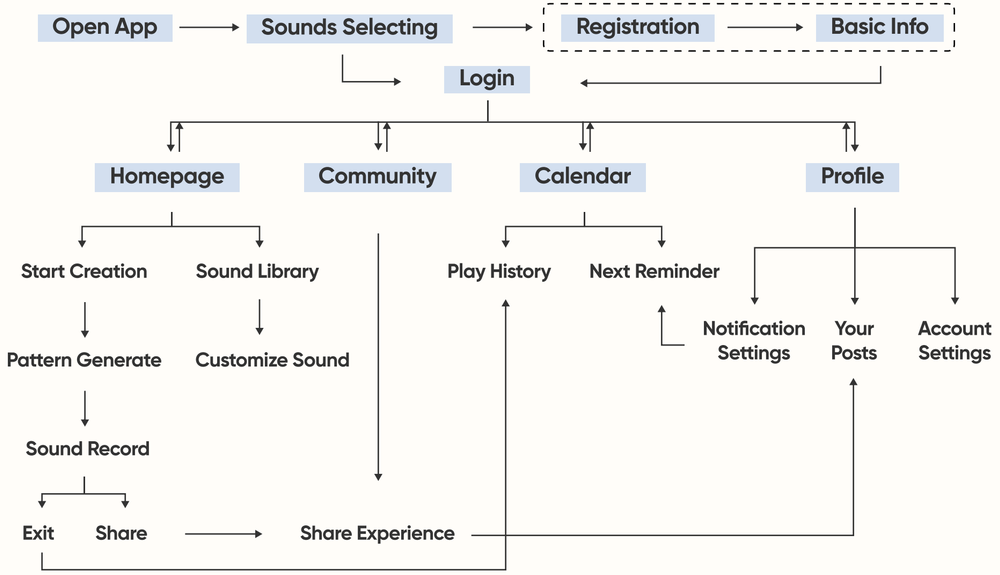
Sounds Selecting
Find the most suitable sound for kid at first start
Sounds can be modified later

Build the Pattern
Give instructions when sound feedback disappear(RRBs appear)

2D to 3D conversion
The Pattern will show in 2D. The bricks can be built in different row but keep the same front pattern.

Reminder for regular intervene
APP notification reminder for strengthen the pairing

Flow Chart



Colors

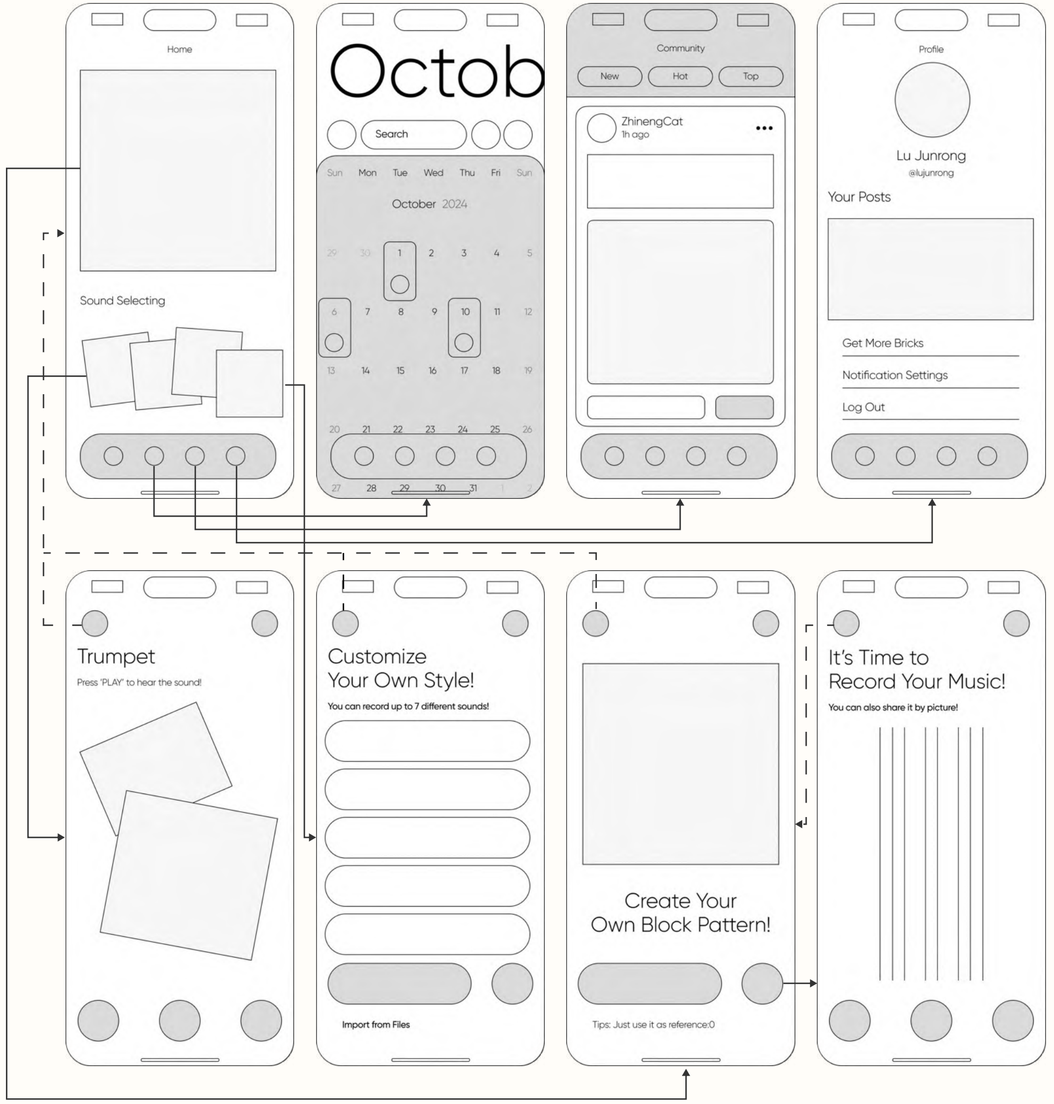
Typology

Regular
Gilory ABCDE

Thin
Gilory ABCDE

Buttons

Wireframes



SoundBricks APP

An innovative APP to facilitate communication between parents and ASD children

The First Test

The first user test, I let the kids tried to build the pattern on their own. And I also asked the teacher to help them if necessary.

Conclusion - 1

Attractive **Flexible**

"The toy is great! It looks interesting!" "Let me try!"

"We never have this kind of toy before" "It fits several classes"

"It combine color and shape recognition, also exercise hands "

Too hard 9 ASD kids from 4-20 tried, not every one of them is able to finish the pattern.

RRBs not happened RRBs like stacking bricks in same direction didn't happen.

Sound is ignored The kids were attractive by the pattern, but ignoring the sound.

Revise - 1

User Assessment The target kids should at least possess skills below:

Cognitive Skills:

Exhibits good sensory-motor skills.

Can perform matching tasks involving three-dimensional shapes and pictures in visual perception.

Possesses some ability to recognize objects (familiar people and actions).

Social and Emotional Skills:

Able to imitate single behaviors.

Shows non-verbal emotional expressions but lacks the ability to name and recognize emotions.

Engages in limited play behavior, with minimal functional play.

Communication Skills:

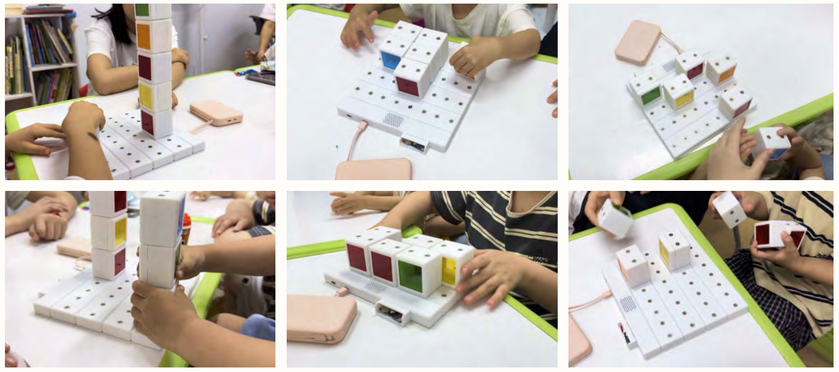
Capable of making requests related to needs and responding to needs-related questions.

Can follow a single instruction, understands "wait a moment," and can answer yes or no questions.

Feng Hua et al. Development-oriented Behavior assessment system for autistic children [M]. Chongqing: Chongqing University Press.2018.

Sound Test **Reduced Shape** Not giving pattern and only use one kind of brick to test sound and let RRBs happen in next test.

The Second Test



The second test focused on sound, so pattern was removed. 5 kinds of sounds is used - piano, drum, saxophone, mechanical and water sound. Every kids will play for a while and I asked them about which sound they like and why.

Conclusion - 2

Sound is noticed The kids noticed the sounds and were able to comment about them.

Preference not clear Some kids didn't have a clear expression about which sound they like most.

Piano sound mostly mentioned Many kids mentioned piano sound. The teacher explained that probably because they both have piano lesson before (the sound is familiar).

Revise - 2

Sound customize is necessary

Customized sound will be a better stimulus for intervention.

Triangle act as RRBs interference

Triangle not allow vertical build after it. So it can also interfere RRBs.

A set of Soundbricks is given to the institute as gift. May every kid grow strong and healthy.

SoundBricks

An innovative System using Stimulus-Stimulus Pairing to intervene ASD.

02

FlexBench

A mobile workshop system for design students

Individual Project

Making-Of: https://youtu.be/v_wOtFdhkD8

Brief

For students at the China Academy of Art's School of Design and Innovation, course projects often involve in making physical products or installations. Nowadays, students are expressing more diverse needs regarding the location of workshops and the management of equipment at the school.

Key Words

Mobile Workshop | Modular Design | Service Design

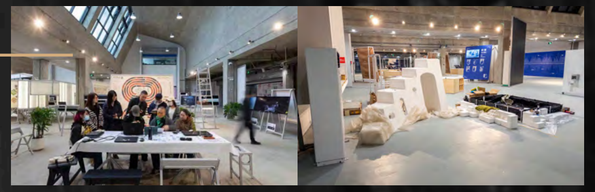
Ideation & Concept

FlexBench is a mobile workshop system serves the needs of SDI's students. Users can borrow workbenches and tools by making online reservations and submitting their needs. To accommodate various spaces and the need for rapid deployment, the system is designed in a modular way. FlexBench can be quickly assembled and freely combined to meet the diverse needs of students working on different types of projects. This system aims to improve the conditions for students when working on their projects.

Background

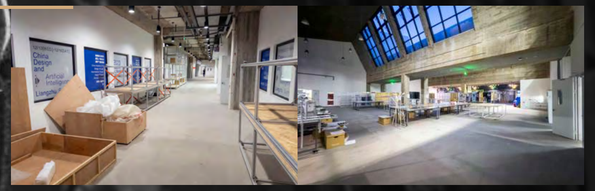
The Current Situation About Making Physical Works

The Liangzhu Campus of the China Academy of Art is a new campus dedicated to design schools. Since the campus has only recently been put into use, there are still shortcomings in the workshop's software and hardware. Students have encountered difficulties while trying to address these issues.



Conflicts in Space Usage

Due to the large size of some projects, also the production process often create noise and strong odors. As a result, many students choose to work in public spaces, such as the corridors outside classrooms or the lobby of Building 8.



Security evicts students

However, since these public spaces serve other purposes, security sometimes asks students to leave when they are working on their projects there.

Unavailable tool system

The school's tool room has not been open to students for a long time due to management issues.

(1)The classrooms for courses change each semester. More than one course have physical outcomes at the same time.

(2)The classrooms are not suitable for making physical projects.

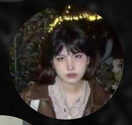
(3)The majority of courses requiring physical outcome are offered by the School of Design and Innovation.

Students & Tutor Interview

Students

"Empty classrooms not scheduled each semester may be occupied by other students, leading to a first-come, first-served situation for using these spaces." **Social "rules"**

"Although classrooms are public spaces, once someone occupies them, you can't stay there."



"I prefer working in my dorm room; a private workspace feels better." **Privacy**

"The current designated activity spaces at the school are too small and overcrowded." **Crowded space**

"Workshops with more tools make it easier to work for long periods and create a better working atmosphere." **Working atmosphere**

"Currently, I mainly work on projects in classrooms, occasionally using the workshop; however, I actually prefer workshop."

"Working on installations in classrooms can be noisy and messy, which is inconvenient." **Convenience**

Tutor

"There aren't many workshop instructors at the school, and not all students know how to use the equipment. When students need to process materials, I can only suggest they go to external factories." **Current system failed**

User Analysis

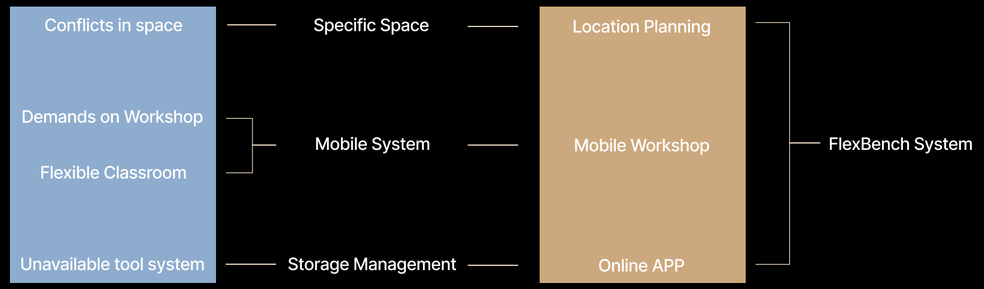
School: School of Design and Innovation

Course: Outcome include installation or physical product

Paint Points:

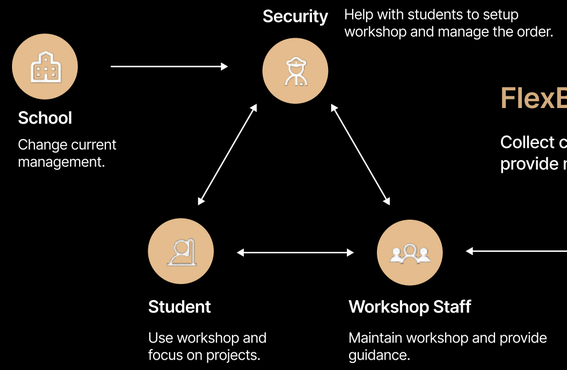
- No space for freely using tools
- Not enough room to work
- Classrooms are overcrowded
- No specific space for work
- Unavailable tool system
- Unable to borrow tools
- Current classrooms are unsuitable
- Working in classroom can affect others
- Classroom often change
- Using public space is not allowed
- Workshop is too far away

Ideation



From the students' perspective, when working on projects, they desire a dedicated space that allows them to create more efficiently and conveniently.

Considering the campus's current situation, where fixed spaces are limited, it would be beneficial to utilize some public areas and adapt them with mobile setups in line with the characteristics of the course schedules.



FlexBench System

Collect course information and student needs to provide mobile workshop service and tools guidance.

FlexBench APP

Student: Check workshop location and ask for help.

Teacher: Book workshop and check workshop status.

Staff: Check workshop status and obtain user's needs.

Workbench System

A modular workbench system suitable for various types of work.

Work Flow Analysis

For a physical project, its work flow might include the following stages involved in different kinds of work that need workshop.

1. Low-Fi Prototyping

Cardboard/Plastic Prototyping

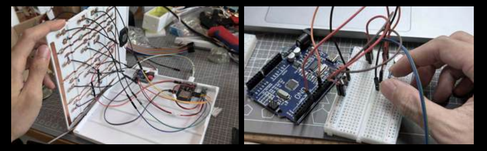
Space Demand: Low

Tools Demand: Low

Workshop Needed: Depend on Size

Large Size: **Workshop**

Small Size: **Classroom**



2. Electronic Prototyping

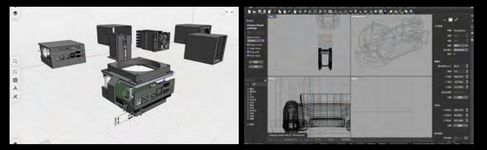
Coding, Soldering & Wiring

Space Demand: Low

Tools Demand: Low

Workshop Needed: No

Best Place: **Classroom**



3. Digital Prototyping

Modelling, Simulation

Space Demand: Low

Tools Demand: Low

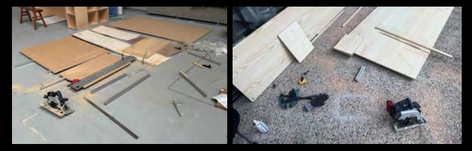
Workshop Needed: No

Best Place: **Classroom**

4. Outside Parts Processing

Outside Factory Processing, Digital Fabrication

Outside



5. Self Processing / Hi-Fi Prototyping

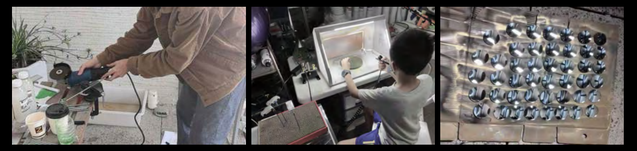
Parts Making(Wood/Metal) & Parts Assembling

Space Demand: High

Tools Demand: High

Workshop Needed: Yes

Best Place: **Workshop**



6. Final Finishing

Sanding & Painting

Space Demand: High

Tools Demand: High

Workshop Needed: Yes

Best Place: **Workshop**

Step 1 could happened in both classroom and workshop depend on the size of the prototype. Steps 2-3 are mostly conducted in the classroom, and from the perspective of space and tool requirements, there are no significant conflicts. For large Low-Fi prototype and steps 5, 6 require more space and tools, and due to current issues with space and management, they need to be carried out in the workshop.

Spatial Analysis

The entire campus can be broadly divided into five types of spaces. Using the area of the School of Design and Innovation as an example, I will analyze the types of spaces and identify suitable locations for setting up workshops.

Central Hall

Lager indoor space, but sometime unavailable for workshop if exhibition held.

Space: Indoor/Large **Traffic: Low**
Note: sometime unavailable (exhibition/event held)

Suitable For Work: Yes

Low-Fi Prototyping | Large Parts Making | Large Parts Assembling

Classroom

Classrooms are small indoor spaces where courses and project presentations take place.

Space: Indoor/Small **Traffic: Low**

Suitable For Work: Yes

Low-Fi Prototyping | Small Parts Making | Small Parts Assembling | Electronic Prototyping | Digital Prototyping

Central Hall Corridor

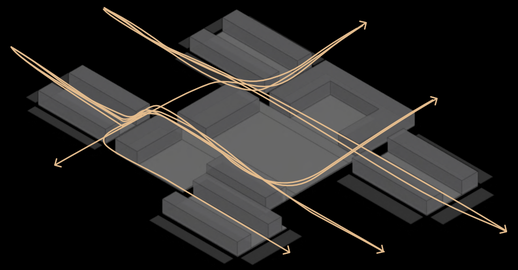
An important passage on campus, but currently, some clutter is being stored there, which somewhat affects traffic flow.

Space: Indoor/Large **Traffic: High**
Note: vital passage of the campus.

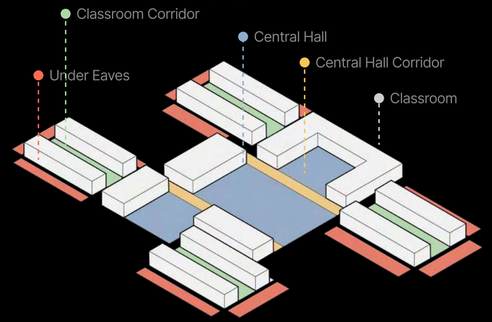
Suitable For Work: No

No Suitable Work Here

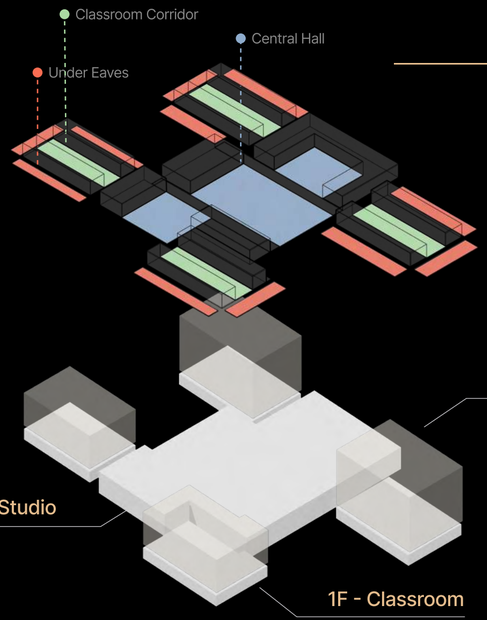
1F Flow



1F View



1F View



Classroom Corridor

The closest spots to classrooms. An indoor space provide shelter for most of the works.

Space: Indoor/Medium **Traffic: Medium**

Suitable For Work: Yes

Low-Fi Prototyping | Large Parts Making | Large Parts Assembling

Under Eaves

The large eave can provide shelter for non-extreme weather. Close to classroom.

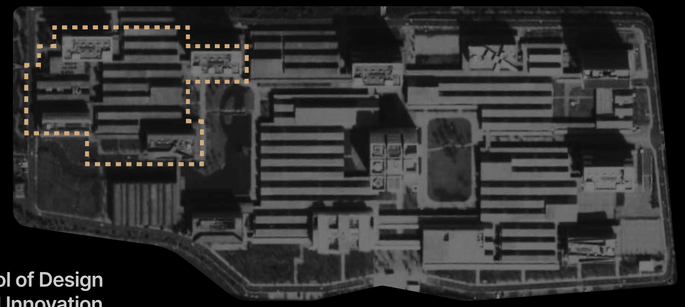
Space: Outdoor/Large **Traffic: Medium**

Suitable For Work: Yes

Large Parts Making | Large Parts Assembling | Final Finishing

Workshop Location

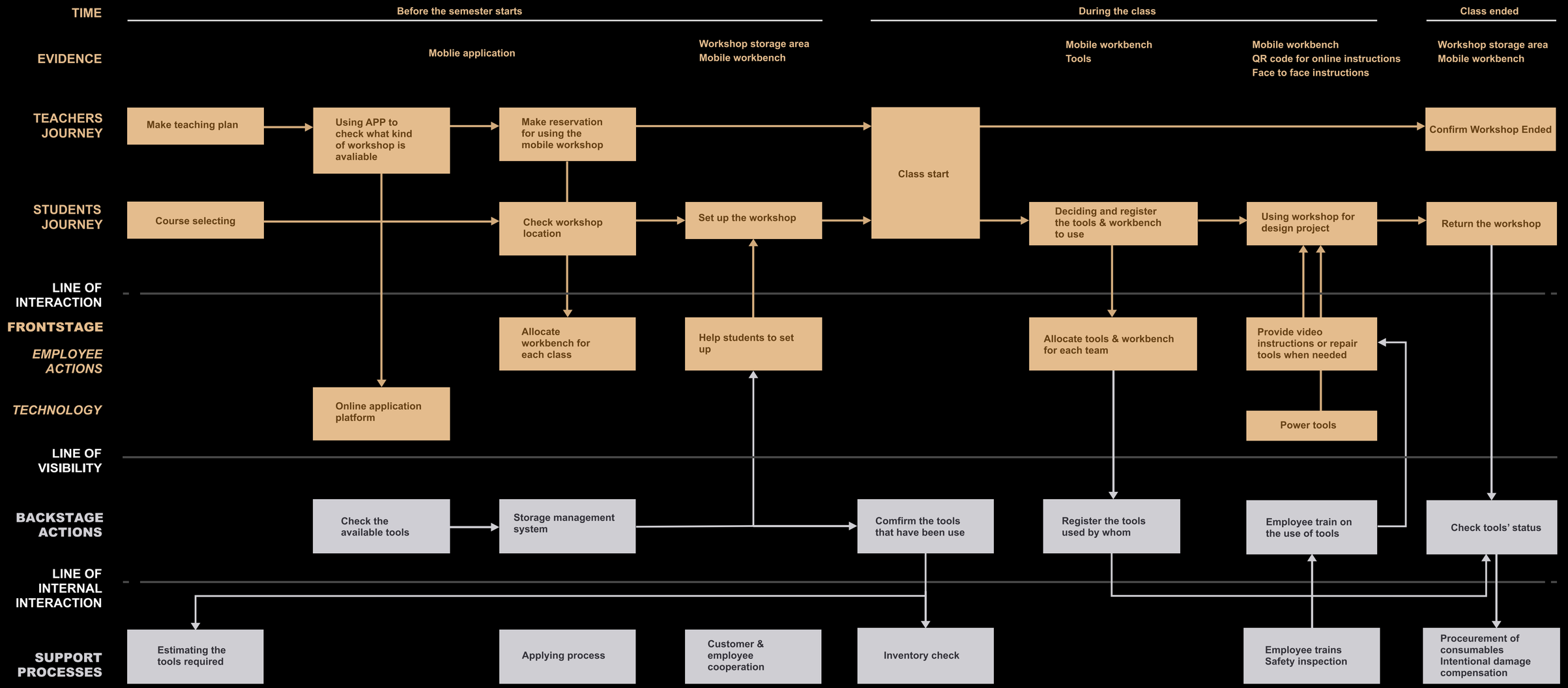
Considering the space size, traffic, and characteristics of the spaces, there are three suitable locations for setting up workshops in the School of Design and Innovation: the central hall, classroom corridors, and under eaves.



School of Design and Innovation

Most buildings share a similar structure, the workshop layout can be extended to other schools.

Service Blueprint



Tools Organizing

1. Power supply

2. Small Hand tool set

3. Light duty fastening set

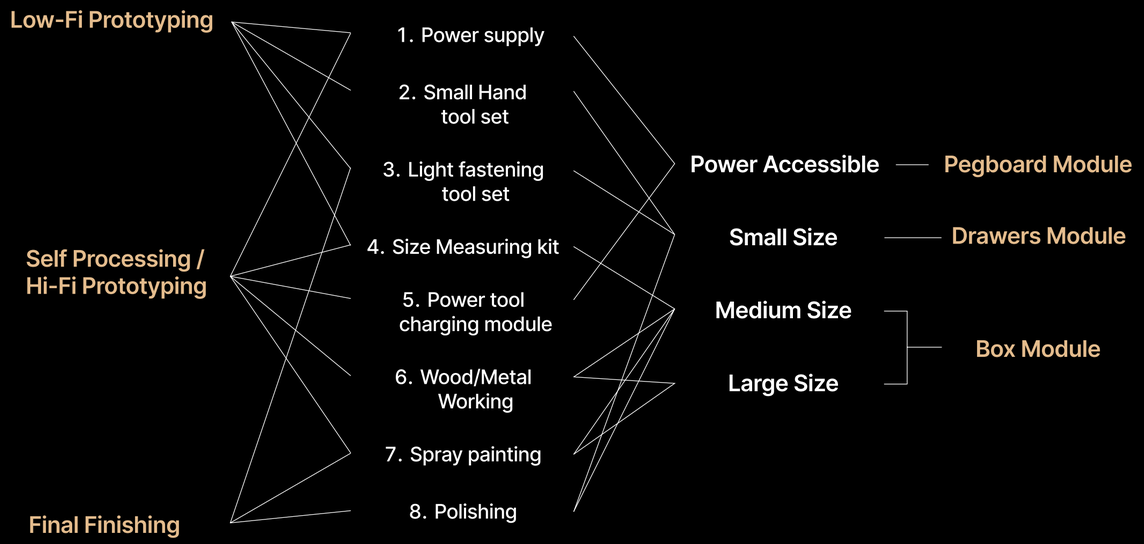
4. Size Measuring kit

5. Power tool charging module

6. Wood/Metal Working

7. Spray painting

8. Polishing

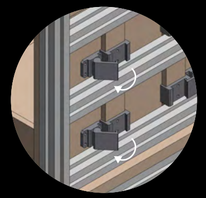


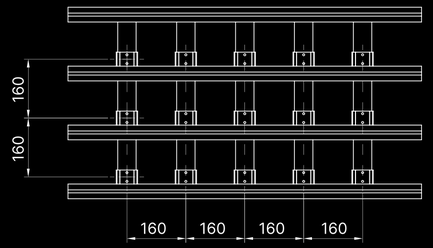
Workbench Design

Snap-fit System

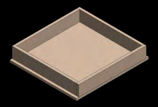
The snap-fit system is used to mount modules.

Center-to-center spacing of 160mm both vertically and horizontally.



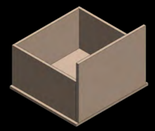


Drawer Module



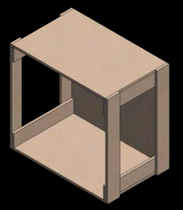
3 Sizes Drawers

Interchangeable for different tools.



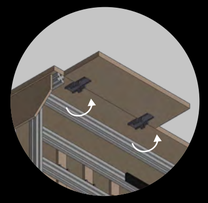
One drawer unit is 80mm height.

Box Module



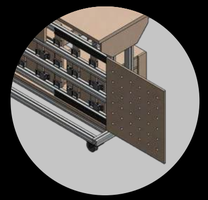
Box module is more flexible on its size to hold large tools.

The tools in the box module are easily accessible.

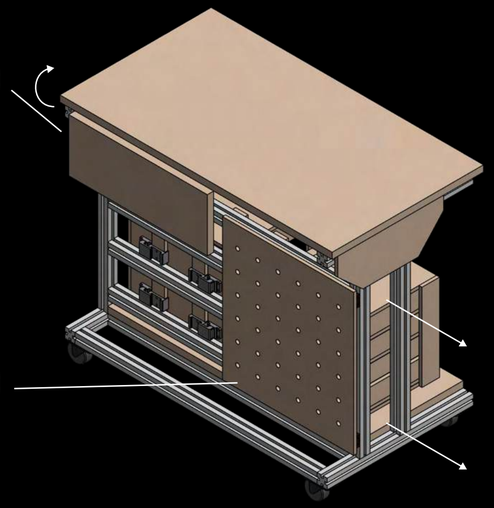


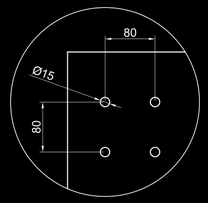
Foldable Extension

Extend the size of the workbench.

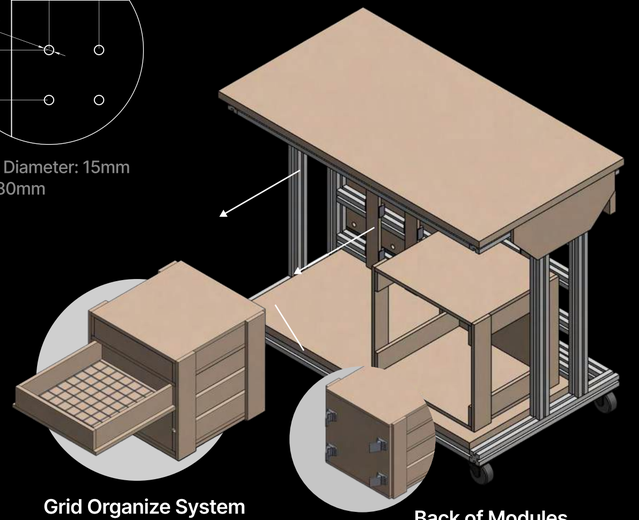


Moveable pegboard





Holes Diameter: 15mm
Gap: 80mm



Grid Organize System

Grid Unit Size: 35mm

Back of Modules

Workbench Present

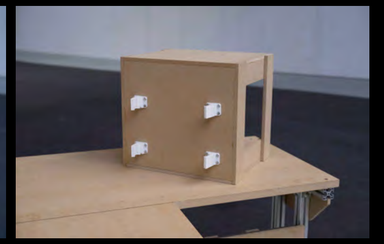
Snap-fit System

Drawers Module - Grid System

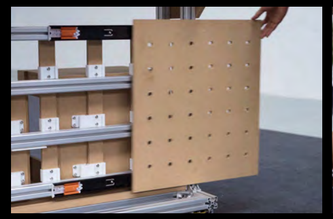
Drawers are interchangeable for different situation.

Holders can be modified with 3D print parts.

Box Module



Pegboard Module



The pegboard module is primarily used for connecting power sources or hang currently using tools.

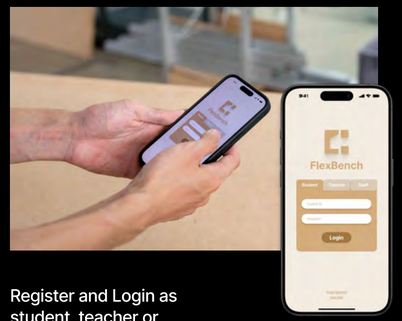
Foldable Extension

The extension can be folded or laid flat and can assist with drilling or cutting tasks.

Story Board

1. Prepare for Classes

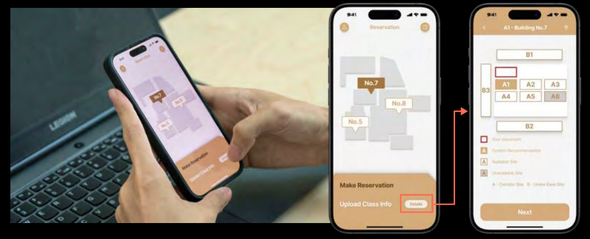
Register & Login



Register and Login as student, teacher or staff.

Teacher:

Login to upload class info and make reservation.



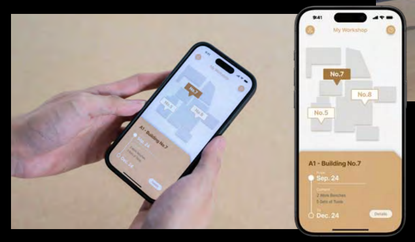
Staff:

Check tools status and confirm reservation.

2. Setup Workshop

Student & Staff:

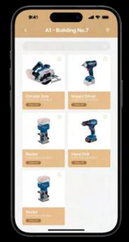
Check workshop location and setup workshop.



Claim Tools

Student:

Claim tools and check guidance.

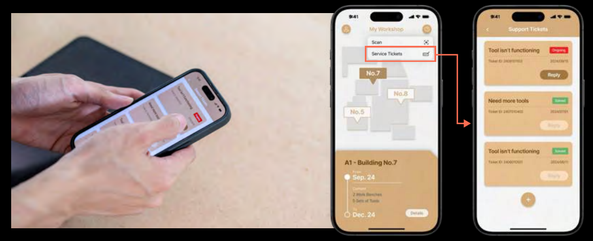


3. Use Workshop For Projects

Reach For Help

Student / Teacher:

Ask for help on Service Tickets page.



Staff:

Provide info or do repair.

4. Return Workshop

Staff:

Check workshop status and count tools.

04

Matrix - Emergence Necessity

An Interactive Installation Design

Individual Project

Demo Video: https://youtu.be/t3h_mF45gQY

Brief

This project explores the dynamic relationship between structure, interaction, and emergence through a modular, matrix-based installation. The design is grounded in the concept of simple components that form complex behaviors when arranged in a matrix and interact with external stimuli. The installation responds to the proximity and movement of participants, demonstrating the potential of emergent behaviors through physical interactions.

Key Words

Matrix Theory | Emergence Theory | Interactive Installation

Ideation

The installation adopts a modular triangular design, with each module capable of interconnecting and infinitely extending to form a matrix. Each module contains three nodes, and each node controls tensioned lines that create a woven effect during movement. Each node is limited to motion in a single direction, but the combination of these simple nodes generates complex motions, embodying the concept of emergence.

The Origin of Matrix

Mathematical Matrix As Structural Matrix

Array Methods

10th–2nd century BCE

The Nine Chapters on the Mathematical Art

Scholars - China

the first example of the use of array methods to solve simultaneous equations, including the concept of determinants.

1545

Ars Magna

Gerolamo Cardano - Italia

Introduced the method to Europe when he published Ars Magna.

Determinants

1683

Development in Japan

Seki Takakazu used the same array methods to solve simultaneous equations

1700

Publicized of Arrays

Gottfried Wilhelm Leibniz publicized the use of arrays for recording information or solutions and experimented with over 50 different systems of arrays.

Matrix

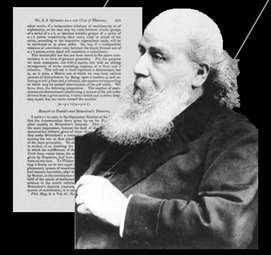
$$\mathbf{A} = \begin{bmatrix} -1.3 & 0.6 \\ 20.4 & 5.5 \\ 9.7 & -6.2 \end{bmatrix}.$$

1850

The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science

37: 363-370 - James Joseph Sylvester - English

The term "matrix" was coined by James Joseph Sylvester, who understood a matrix as an object.



Matrix as Structure

Matrix as Organization

Matrix as Form

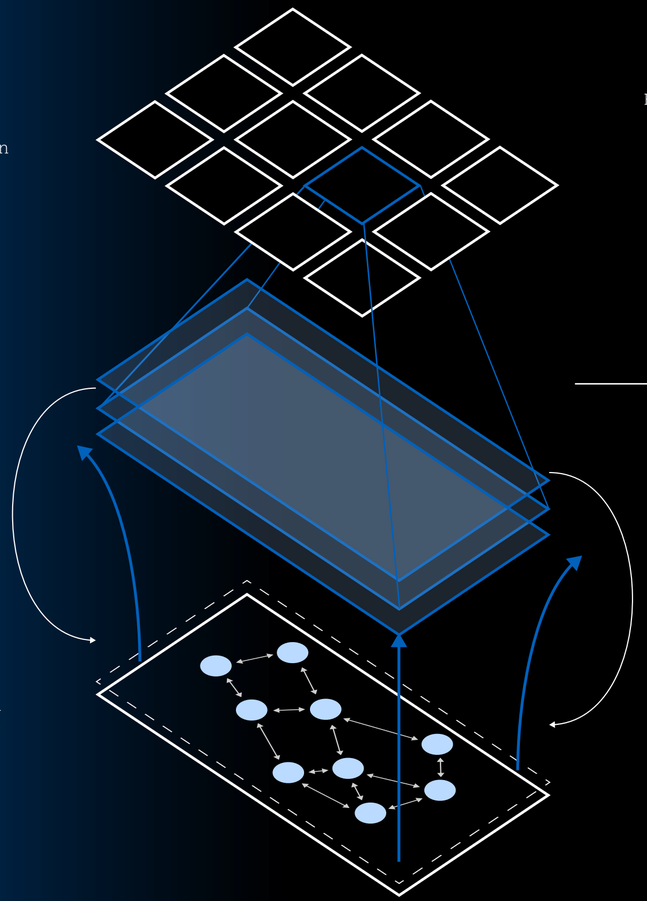
A matrix, at its core, is a structured arrangement of data or elements, typically organized in rows and columns.

Emergence In Matrix

Matrix As Construct, Emergence As Result

Muti-Layer System Model

Fromm, J. (2005). Ten Questions about Emergence. arXiv: Adaptation and Self-Organizing Systems. (revised by me)



System

A System that contain many parts.

Sub-System

Sub-system could be multi-layered.

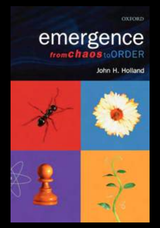
Individuals

The minimal layer of the system.

Higher Level

Lower Level

How Matrix System To Be Emergent?



Emergence: From Chaos to Order

Holland, John H, Emergence From Chaos to Order (Oxford, 1998; online edn, Oxford Academic, 31 Oct.

Networks of relationships constitute wholes with characteristics that cannot be reduced to those of their components. Every subsequent level of emergence is, therefore, more than the sum total of its components and follows its own regularities and contingencies.

New Characteristics

An emergent system exhibits new characteristics that a simple collection of individual elements cannot display.

Emergence

Simple local interactions within a system generate complex overall behavior.

Matrix

- Individual interaction
- Feedback mechanism
- Self-organized

Individuals form a matrix through all this structured relationships by arranging and connecting in an self-organized pattern.

Form New System As Individual

Ideation

The installation aims to simulate the formation of emergence. A network of interactions at the individual level forms a matrix both structurally and systematically, from which characteristics not present at the individual level emerge.

Human

Act as outside stimulus to the system

Units Interaction

React to inside and outside stimulus

The System

Individual Unit

Trianglar Shape

Use simple forms and singular motion to simulate an individual lacking complex characteristics.

Matrix

Formed by units in structure

Connected by interaction in organization

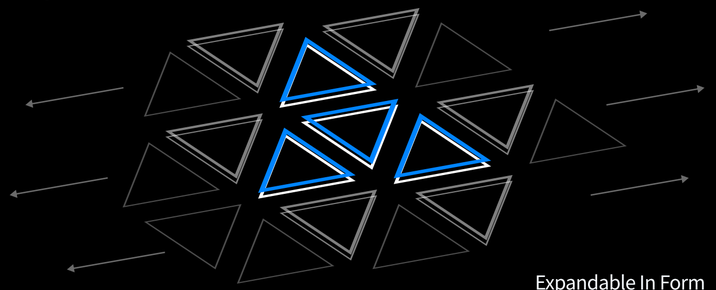
Emergence Necessity

Group behaviors, distinct from the movement of a single unit, will eventually emerge, thus replicating the phenomenon of emergence.

Concept

Structure

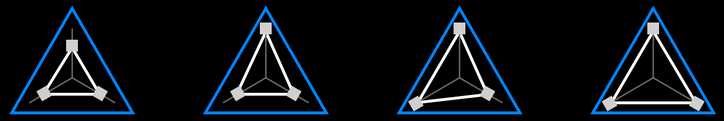
Triangle As Matrix Unit



Expandable In Form

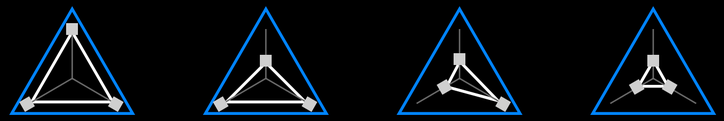
Movement

Expand



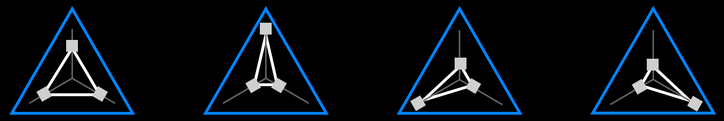
The servo motor drives the node to expand outward.

Contract



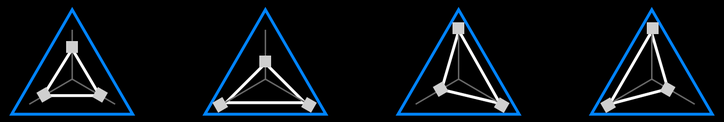
The servo motor drives the node to contract inward.

Approach



The servo motor drives the node to approach the direction of the stimulus.

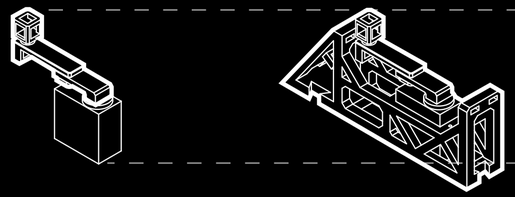
Retreat



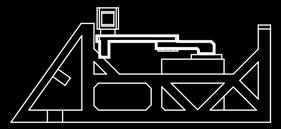
The servo motor drives the node to move away from the direction of the stimulus.

Structure Detail

Node



A single node consists of a frame, connecting rods, and a servo motor, with thin lines pulling between the nodes to enhance the form of movement.



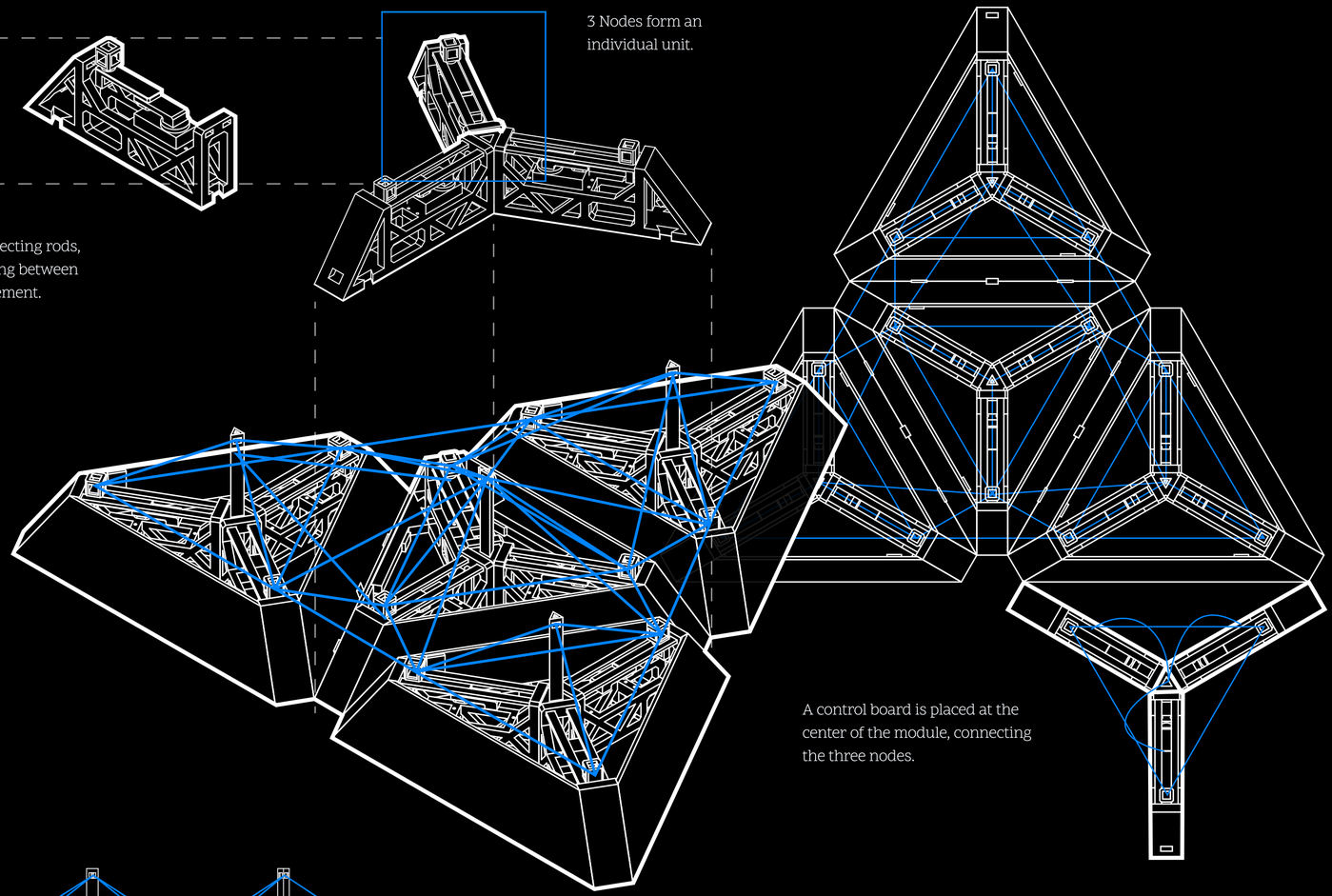
The connecting rod structure converts the rotational motion of the servo motor into linear reciprocating motion.

Individual Unit

3 Nodes form an individual unit.

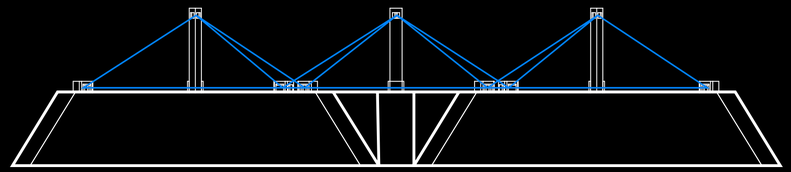
Top View

LED light strips are arranged inside the slanted side panels, allowing for different lighting effects.



A control board is placed at the center of the module, connecting the three nodes.

Side View



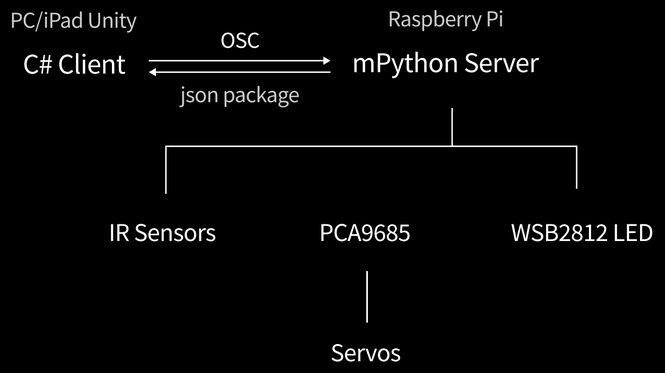
The central pillar of the module pulls lines toward the nodes, creating dynamic motion on the sides.

The lighting and movement will be controlled remotely via LAN >>

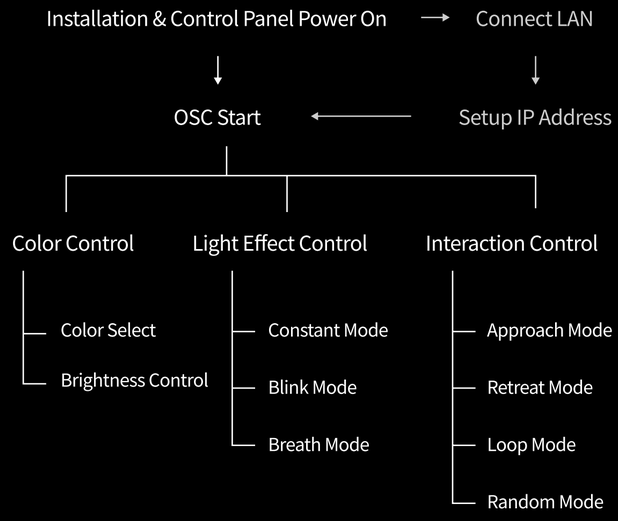
Control Panel

Backstage control panel built by Unity UI kit and Open Sound Control (OSC) to control the installation from PC.

Control Hierarchy



Setup Flow




```
public string id;
public string server;
public int timestamp;
}
public class SimpleMessageTransmitter : MonoBehaviour
{
#region Public Vars {

public string Address = "/jsonmsg";

[Header("OSC Settings")]
public OSCTransmitter Transmitter;

#endregion

public static int timestampcount = 0;

#region Unity Methods
public void sendmessage(string message_command, string message_mode, string message_id)
{
 MessageStart ms = new MessageStart();
 ms.command = message_command;
 ms.mode = message_mode;
 ms.id = message_id;
 ms.server = "triangle";
 ms.timestamp = timestampcount ++;

 string json = JsonUtility.ToJson(ms, true);

 var message = new OSCMessage(Address);
 message.AddValue(OSCValue.String(json));
 Transmitter.Send(message);
 print(json);
}

protected virtual void Start()
{
 sendmessage("conn", " ", "apollooo1");
}
#endregion
```

Light Color Control

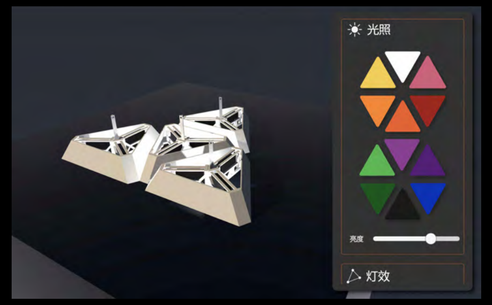
The lighting offers 12 color options, with a slider below to adjust brightness.

Light Effect Control

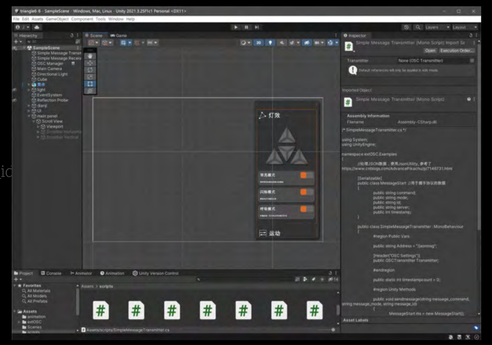
There are three lighting effects to choose. Different lighting effects can be paired with various interaction modes.

Interaction Control

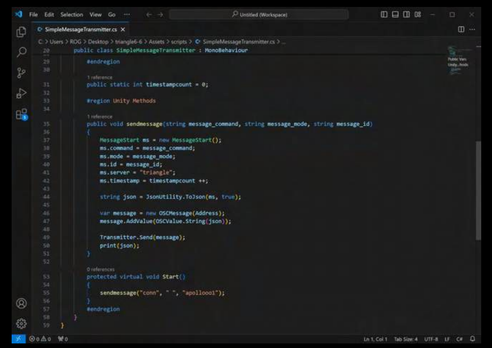
Four motion modes, two of which are interactive. The other two movements are not affected by external stimuli.



Control Panel Screenshot



Unity UI development

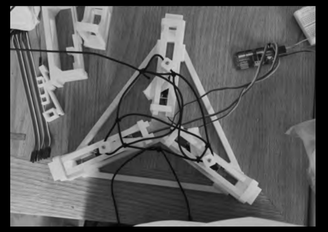


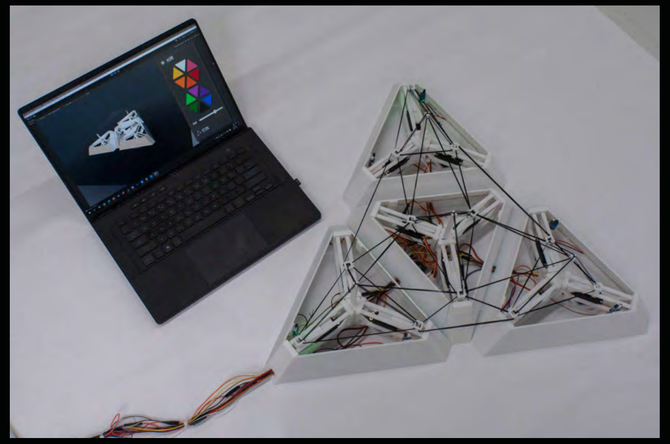
C# OSC Transmitter

Installation & Control Panel

Make Process

Use 3D printing to iterate and test the effect of different lines.





Final Present

Demo Video: https://youtu.be/t3h_mF45gQY

I created a 4-unit demo to showcase the design, with each unit being fully modular and capable of expanding in any quantity or configuration.

Although each node's movement is a simple back-and-forth motion, the interactive movements display a dynamic whole that individual nodes alone cannot achieve, with emergence being replicated through the matrix-like motion.

猫友记Pawpals

每一条尾巴都有一个故事。Every tail has a tale.

毕业设计作品

猫友迹 PAWPALS

作品概述

《猫友迹》是一套基于多模GNSS定位技术的流浪猫管理服务系统，集成了猫用穿戴式定位设备、IoT流浪猫站点与多端APP，为城市流浪猫问题提供数据驱动的系统性解决方案。其创新点在于将IoT技术、流浪猫关怀、社区协作三者融合，在行为方法论上结合非人类中心主义与共生理念，体现对动物权益的尊重，打破传统依赖人工观察的碎片化模式，构建出可持续、系统化的TNR支持平台。

选题由来

1 我们选择的物种 -- 流浪猫

截至 2021 年，中国约有 5 300 万只流浪猫。据估计，美国的流浪猫数量约为 3,000 万至 4,000 万只。这些数字受到宠物遗弃、宠物走失和猫的高繁殖率等因素的影响。

种群特点

数量多

捕猎野生动物

捕捉和管理难度较大

繁殖能力极强

适应性强 频繁在人类区活动

领地意识

1、尿液、信息素、抓痕标记

2、对领地进行巡逻、防御

3、对熟悉环境的依赖

共享领地

1、资源充足

2、建立社会等级并共享领地

3、人类的影响（多猫家庭）

智能猫

公猫（已绝孕） 1岁半

2 流浪猫面临的危险

资源匮乏

没有稳定的食物和水源,营养不良会导致它们寿命减少，体弱多病。

恶劣天气

流浪猫常常无法在恶劣的天气下找到避难所。在冬天，流浪猫的冻死率大概在60%

虐猫

每年有近100万只动物在暴力事件中被虐待或杀死，对动物的虐待和忽视在农村和城市都很普遍。

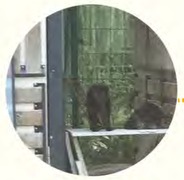
无人看管

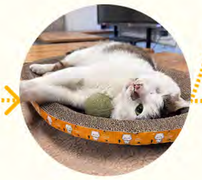
缺乏日常喂养和看护，受伤后无法及时医治。

媒婆

暑假失去了一只眼睛

现有的流浪猫干涉流程 ——以“有良犬”为例


发现流浪猫


固定喂养

医治

TNR绝育 人道且有效的控制流浪猫数量的方法。 Trap Neuter Return

抓捕 → 绝育 → 放归

- 只捕杀会破坏生态系统（麦夸里岛案例）
- 真空效应

赞助/募捐

售卖周边

参与线下活动

制作周边

领养

该流程特点

基于学生社群进行维护

小范围相对完善的系统

能够起一定的引导作用

1 为什么TNR有效

1、控制种群数量

2、绝育后激素平稳，减少攻击性

3、避免真空效应，维持生态平衡

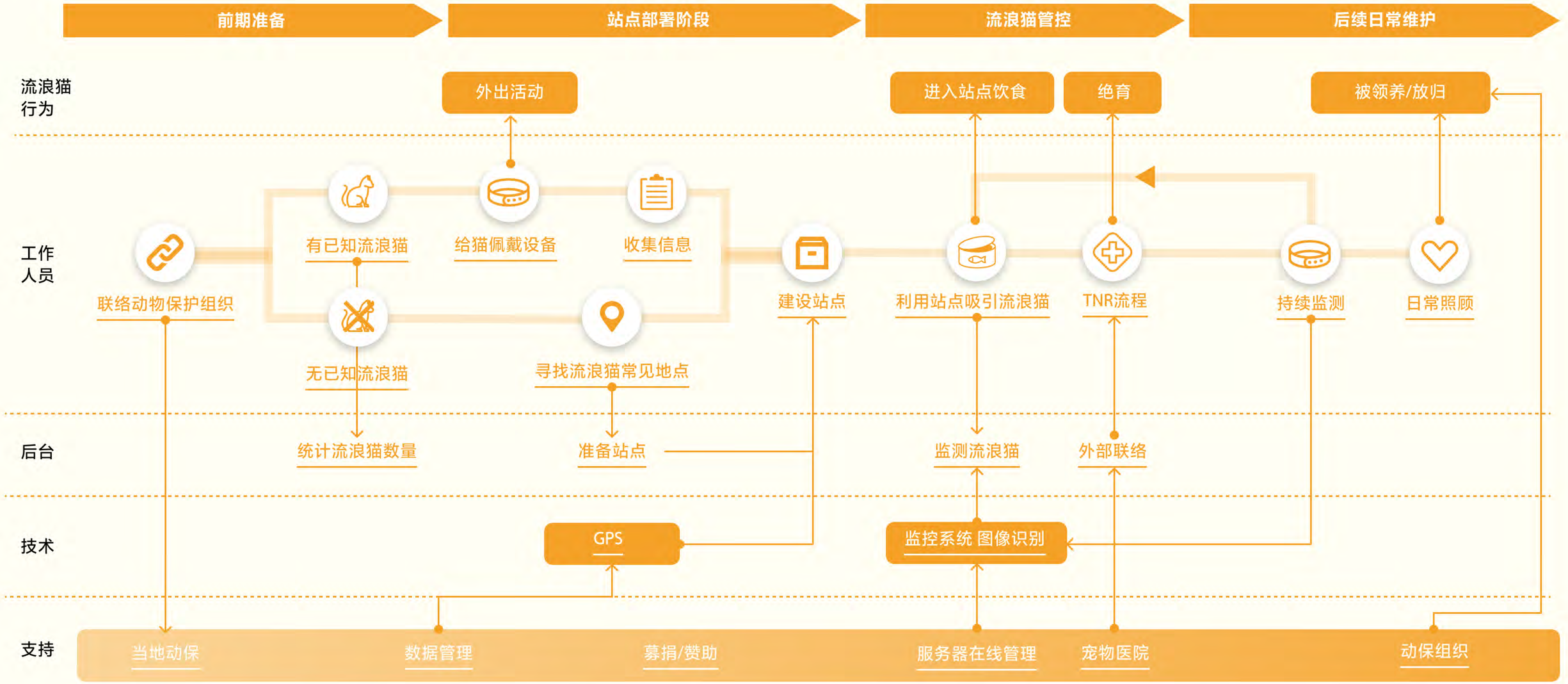
2 结合TNR的发现流程

1、通过已知流浪猫的行为模式推测新的流浪猫可能出现的位置

2、弥补TNR流程缺失的发现环节

3、尝试建立一个可持续的系统

服务蓝图


前期准备
站点部署阶段
流浪猫管控
后续日常维护
流浪猫行为
工作人员
后台
技术
支持
外出活动
进入站点饮食
绝育
被领养/放归
有已知流浪猫
给猫佩戴设备
收集信息
联络动物保护组织
建设站点
利用站点吸引流浪猫
TNR流程
持续监测
日常照顾
无已知流浪猫
寻找流浪猫常见地点
统计流浪猫数量
准备站点
监测流浪猫
外部联络
GPS
监控系统 图像识别
当地动保
数据管理
募捐/赞助
服务器在线管理
宠物医院
动保组织

利益相关者地图 ——以“星座”隐喻*作为补充

项目中理论概念对应

MTHD	星	星座
OOO	对象	
ANT	行动者	网络
猫友记	猫/人/设计物	设计系统

*Paul Coulton & Joseph Galen Lindley (2019): More-Than Human Centred Design: Considering Other Things, The Design Journal, DOI: 10.1080/14606925.2019.1614320

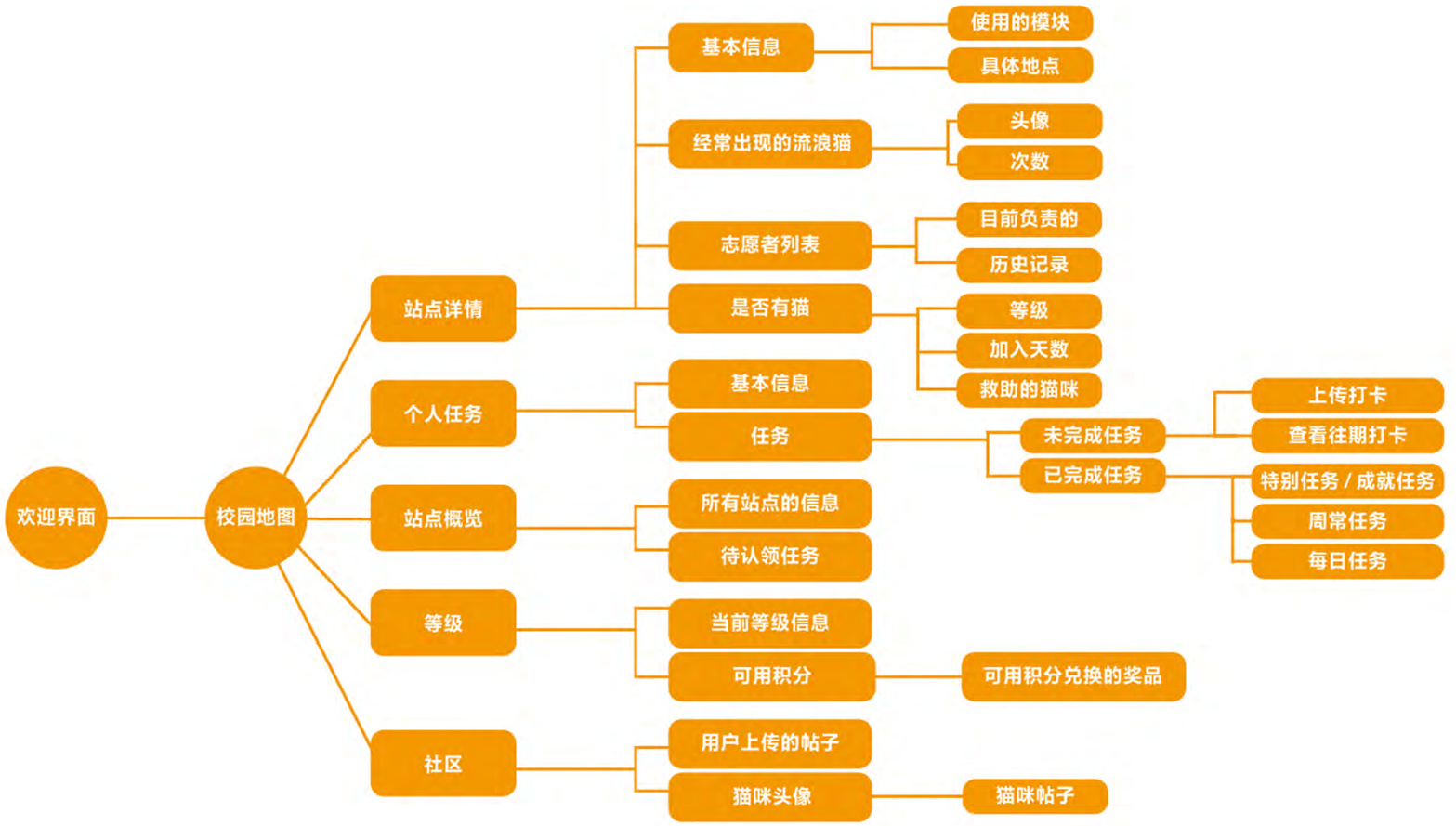
系统构建

每一条尾巴都有一个故事。Every tail has a tale.

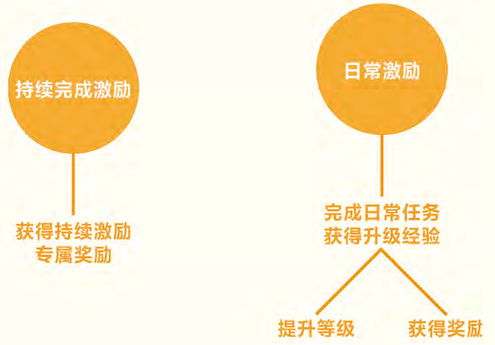
用户旅程构建

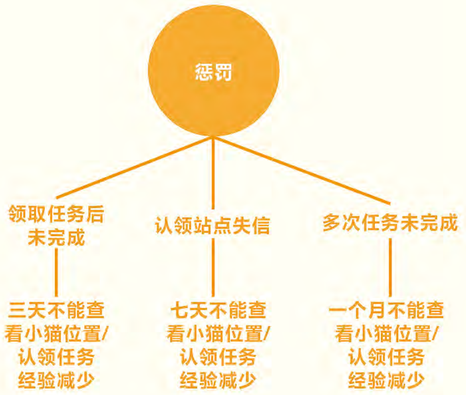
前期准备
站点部署阶段
流浪猫管控
后续日常维护
想了解猫咪近况的人
查看别人分享的帖子
想摸猫的人
打开APP
查看哪些站点有猫
查看站点监控
查看站的具体位置
去摸猫
分享
领取站点的人
达到认领站点的条件
了解认领站点的信息
认领站点
打开APP
查看已经领取的站点详情
完成任务
分享
获得奖励
路过好奇的人
路过站点
扫描站点二维码
查看站点信息
分享

软件平台信息架构 & 用户激励措施

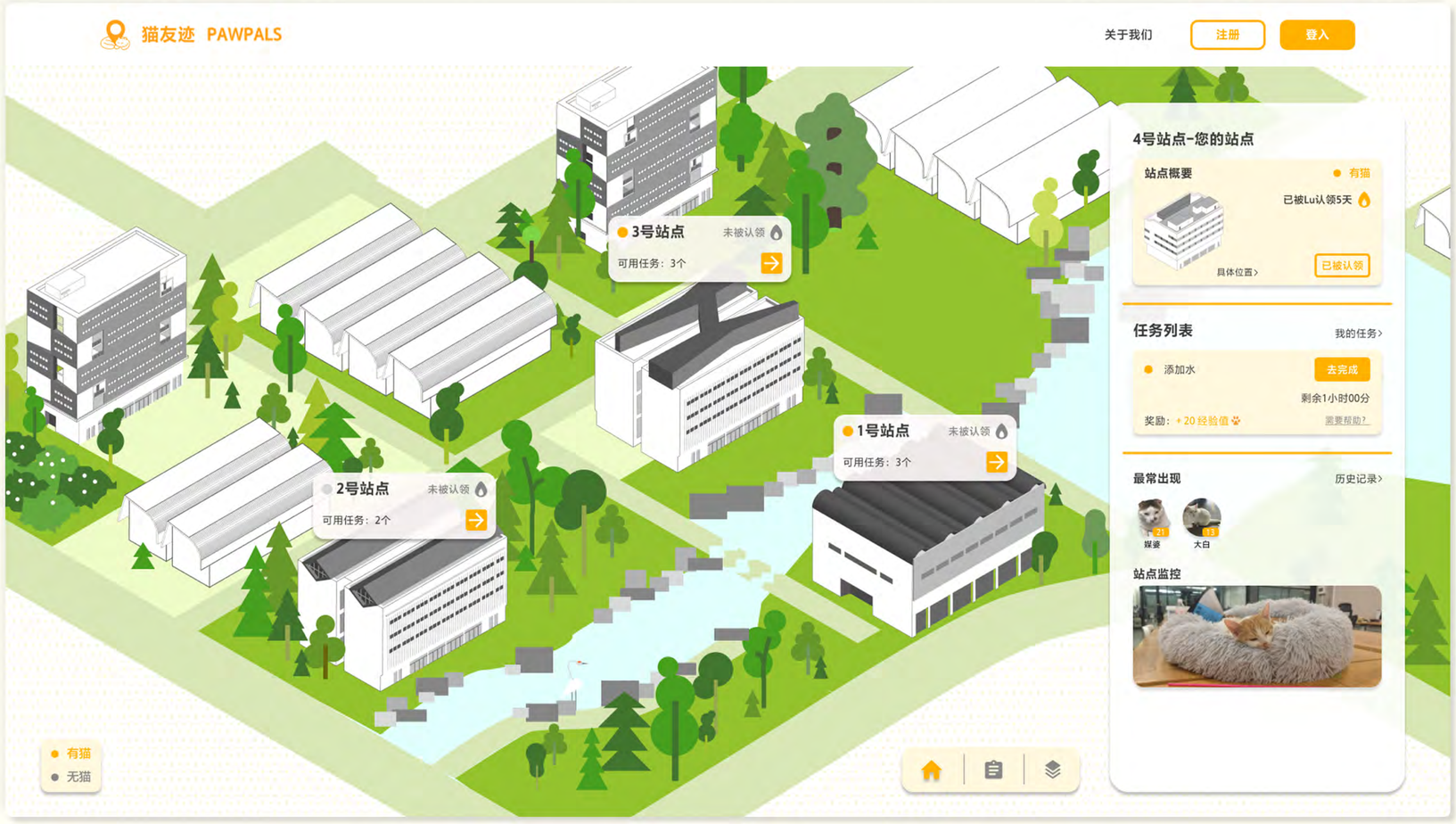


激励措施设计





用户界面


猫友迹 PAWPALS
关于我们
注册
登入
3号站点
未被认领
可用任务：3个
1号站点
未被认领
可用任务：3个
2号站点
未被认领
可用任务：2个
有猫
无猫
4号站点-您的站点
站点概要
有猫
已被Lu认领5天
具体位置>
已被认领
任务列表
我的任务>
添加水
去完成
剩余1小时00分
奖励：+20 经验值
需要帮助?
最常出现
历史记录>
媒婆
大白
站点监控

9:41
3号站点
未被认领
可用任务：3个

9:41
成长中心
Lv 2 投喂小能手
Lv2
Lv3
650
可用积分 1150
全部
可兑换
实物周边
头衔
喵喵吧唧
150 积分
智能猫钥匙扣
150 积分
增长杯子
笔记本

9:41
社区
分享你发现的小猫吧!
最新活动
喵喵
最新活动：今日6:45
小智能
最新活动：前天8:00
猫友空间
这是新来的猫吗?
Tong
小智能太可爱啦！！

9:41
3号站点
站点概要
有猫
未被认领
具体位置>
认领站点
任务列表
我的任务>
今日任务
添加水
+20 经验值
领取任务
添加猫粮
放弃任务
去完成
剩余1小时00分
奖励：+20 经验值
需要帮助?
特殊任务
绝育
+80 经验值
申请领取
最常出现
历史记录>
媒婆
大白

产品原型开发

每一条尾巴都有一个故事。Every tail has a tale.

穿戴设备技术测试

1 GPS测试代码

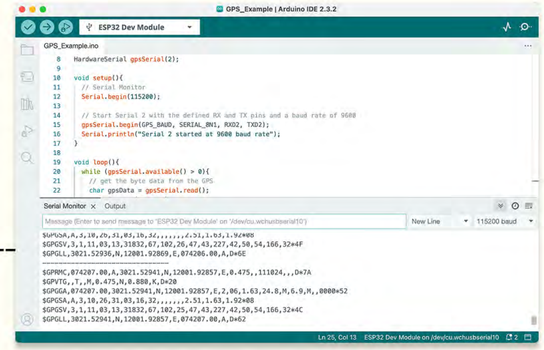
```
// Define the RX and TX pins for Serial 2
#define RXD2 16
#define TXD2 17

#define GPS_BAUD 9600

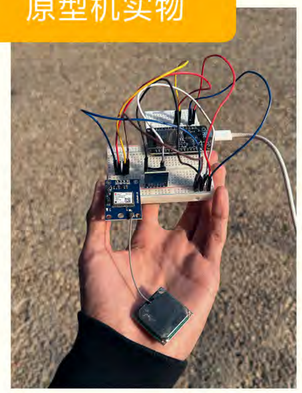
// Create an instance of the HardwareSerial class for Serial 2
HardwareSerial gpsSerial(2);

void setup(){
  // Serial Monitor
  Serial.begin(115200);
  // Start Serial 2 with the defined RX and TX pins and a baud rate of 9600
  gpsSerial.begin(GPS_BAUD, SERIAL_8N1, RXD2, TXD2);
  Serial.println("Serial 2 started at 9600 baud rate");
}

void loop(){
  while (gpsSerial.available() > 0){
    // get the byte data from the GPS
    char gpsData = gpsSerial.read();
    Serial.print(gpsData);
  }
  Serial.println("-------------------------
  delay(1000);
}
```



原型机实物



2 数据分析

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```

$GPRMC,074247.00,A,3021.53039,N,12001.92917,E,0.181,,111024,,,D*72

1. UTC (Coordinated Universal Time)时间,hhmmss（时分秒）格式
2. Latitude，纬度ddmm,mmmm（度分）格式
3. Longitude，经度dddmm.mmmm（度分）格式
4. 地面速率（000.0~999.9节）Knot

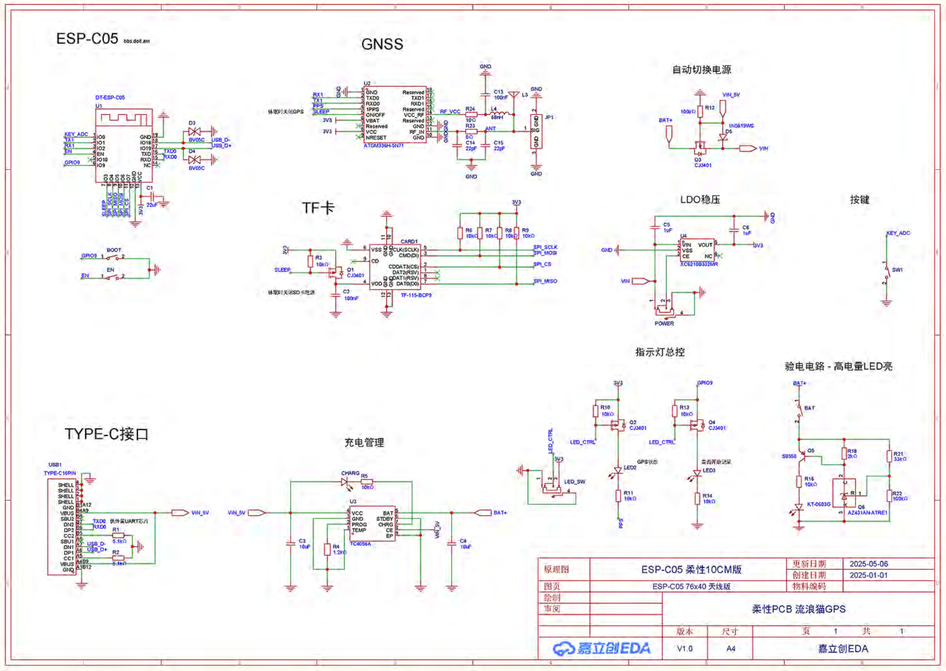
对NMEA数据集三种格式GPGGA(全球定位数据)、GPGSA(卫星PRN数据)、GPRMC(运输定位数据)解析，且本项目中需要可视化的数据有：维度、经度、速度，采用对GPRMC(运输定位数据)进行处理并可视化。

轨迹可视化

硬件选型与算法实现

硬件选型

主控	DOIT(四博智联) ESPC3-05	
GNSS模块	中科微 ATGM336H-5N71	
电源管理	锂电管理 - TC4056A	LDO - XC6210B332MR
数据存储	TF-115模块	



算法实现

系统采用按定时唤醒设置的睡眠机制，结合ESP32-C3休眠状态低功耗特性实施周期性唤醒作业。

下面是休眠部分的关键代码展示：

```
esp_sleep_enable_timer_wakeup(TIME_TO_SLEEP * uS_TO_S_FACTOR); #设置唤醒周期

digitalWrite(STATUS_Pin, LOW); #将外围设备断电

Serial.println("Going to sleep now");

Serial.flush();

esp_deep_sleep_start(); #开始休眠
```

以下是解析数据并写入TF卡的函数：

```
while (millis() - start < 5000) {

while (gpsSerial.available() > 0) {

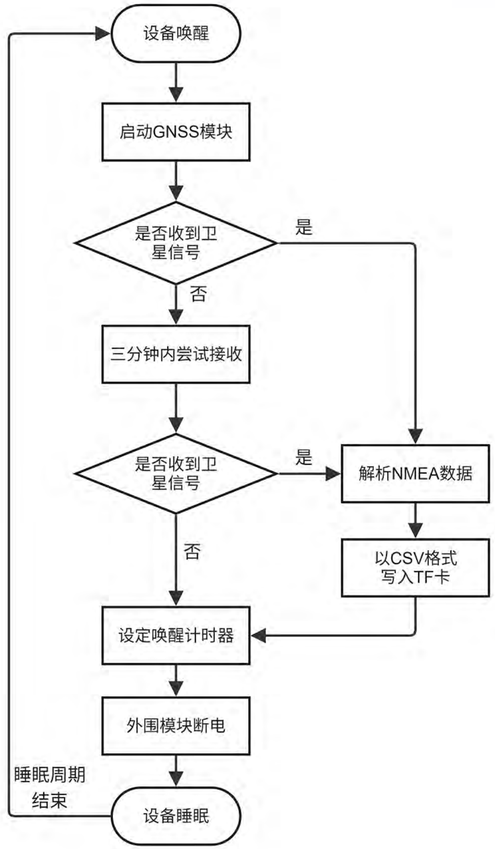
gps.encode(gpsSerial.read());

}

if (gps.location.isUpdated()) {

writeGPS(SD, "/data.txt");

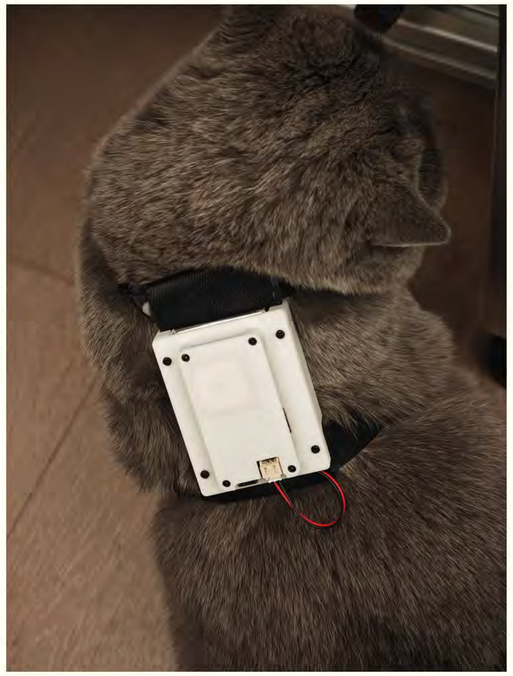
}
```



设备算法实现流程图

穿戴产品设计与用户研究

产品结构与外形设计



初期原型机设计

采用可调节的背包样式。背包绑带用尼龙绑带，通过多段绑带缝制而成，兼顾强度以及舒适度。

柔性项圈实拍图

在结构上，为了提高设备的实际佩戴体验，柔性版本的电路板本身构成了项圈。固定方式上，在项圈的背面设置了魔术贴，便于拆装，能提升流浪猫在佩戴过程中的舒适度和长期佩戴的稳固性。

产品用户测试与反馈

表 3-5 常规 PCB 背包式版本测试结果（表格来源：作者自制）

测试对象	穿戴难度	穿戴后反应	活动受限情况	设备佩戴状态
阿姆	简单	轻微抓挠	观察到走路不稳	良好
兜兜	适中	无异常	走路被绊到	良好
智能猫	难	反抗，逃跑	跑步过程中被绑带捆住	被翻到胸前
媒婆	难	大声叫	不肯走动，并且尝试撕咬	手从项圈中穿出

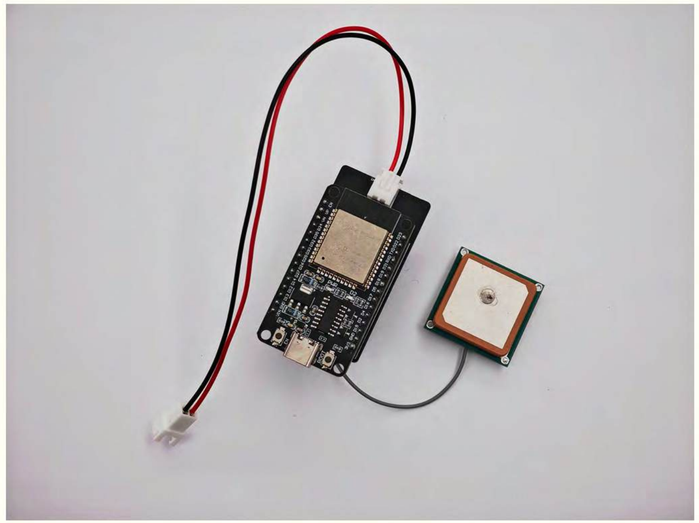
表 3-6 FPC 柔性项圈版本测试结果（表格来源：作者自制）

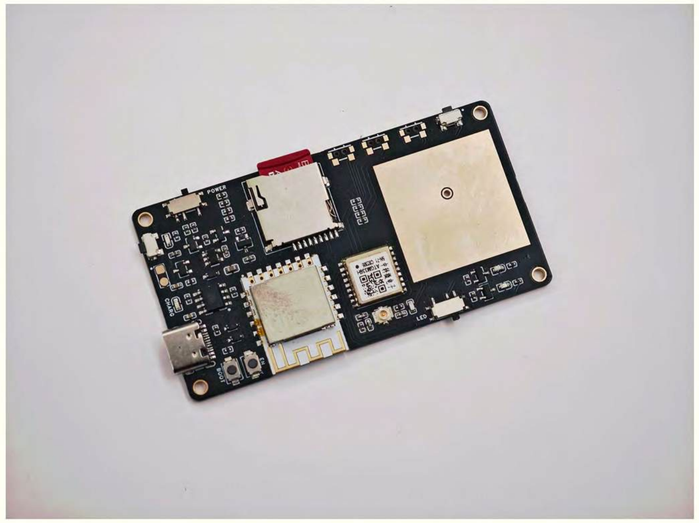
测试对象	穿戴难度	穿戴后反应	活动受限情况	设备自身状况
阿姆	简单	无明显异常	无明显异常	良好
兜兜	简单	轻微抓挠	无明显异常	因体型小略松动
智能猫	适中	叫了数声后适应	无明显异常	良好
媒婆	适中	无明显异常	无明显异常	良好

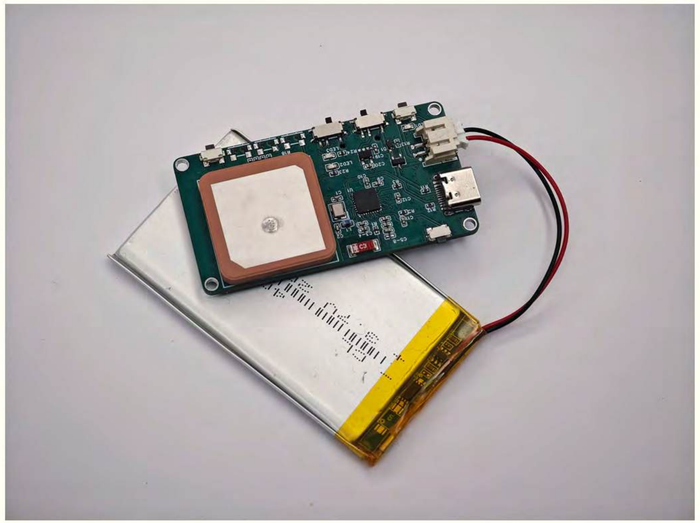
FPC柔性版本测试效果良好，在流浪猫脖子上一套一粘即固定好，固定速度快且牢固。

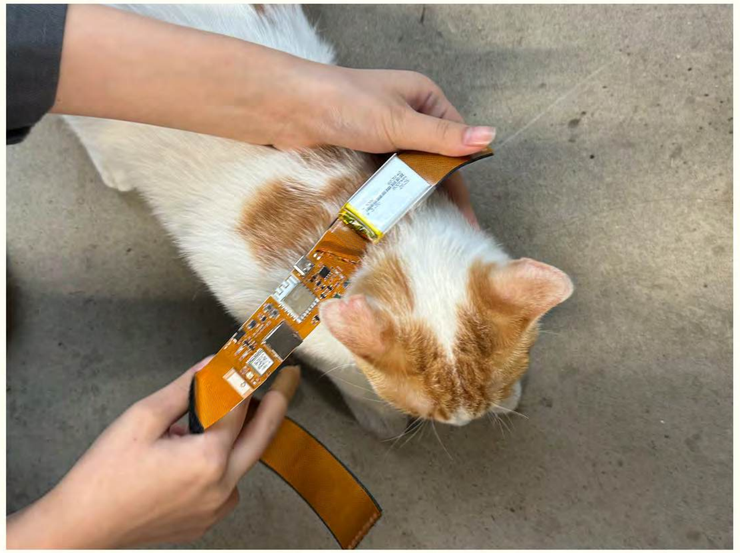
测试中流浪猫的佩戴反馈都达到了目标期望水平。采用柔性电路板以及魔术贴式固定能够有效的减少流浪猫的排斥感以及提升动物保护组织成员的佩戴效率。

测试过程图片









站点搭建

每一条尾巴都有一个故事。Every tail has a tale.

流浪猫行为学与功能分析

流浪猫行为学特征

表 2-2 流浪猫行为类型以及影响因素（表格来源：作者自制）

活动类型	觅食	捕猎	休憩	领地巡逻	猫社交互动
引导因素	食物资源	获取食物、娱乐	空间资源	自身安全	繁殖

行为观察证实，食物的来源分布对猫的活动路径起到决定性影响，它们会围绕固定的食物投放点与人类聚集区展开巡回行动。流浪猫对栖息地的选择偏好安静、隐蔽、少有人打扰的位置，就如楼道的死角区域、废弃构造物或草木茂盛的地带，这些区域往往会成为其昼伏夜出生活节律的关键地点。

模块类型	满足需求	
猫有食	食物资源	
猫有架	空间资源	娱乐
猫有窝	自身安全	休憩
猫有藏	工作人员维护	

主体 猫有塔
- 猫有窝
- 猫有食

扩展
- 猫有架
- 猫有藏

站点设计

站点安置红外感应与摄像头，记录流浪猫的出入情况，而且结合线上平台，实时对流浪猫状态加以更新，方便志愿者和管理人员掌握流浪猫的动态。

基于ESP-NOW协议的远程通讯

传感器及门机构

最终展陈